

#1 · Number

What is a rational number?

Answer

A rational number is any number that can be written as a fraction of two integers, where the bottom number is not zero. This includes all integers, fractions, and decimals that either stop or repeat. For example, 5, $-2/3$, and 0.75 are all rational.

#2 · Number

What is an irrational number? Give an example.

Answer

An irrational number is a decimal that goes on forever without ever repeating a pattern. It cannot be written as a fraction of two integers. Pi (3.14159...) and the square root of 2 are famous examples.

#3 · Number

Is the repeating decimal 0.333... rational?

Answer

Yes. A decimal that repeats a pattern forever is rational because it can be written as a fraction. In this case, 0.333... equals exactly $\frac{1}{3}$.

#4 · Number

Which is smaller: -10 or -2?

Answer

-10 is smaller, because it sits farther left on the number line. Many students think -10 is bigger since 10 is a bigger digit, but the negative sign reverses that. The farther left a number is, the smaller its value.

#5 · Number

How do you compare two fractions like $\frac{3}{4}$ and $\frac{5}{8}$?

Answer

Rewrite them with a common denominator, then compare the numerators. Here $\frac{3}{4}$ becomes $\frac{6}{8}$, and since 6 is bigger than 5, $\frac{3}{4}$ is bigger than $\frac{5}{8}$.

#6 · Number

Which is smaller: $-\frac{1}{3}$ or $-\frac{1}{4}$?

Answer

$-\frac{1}{3}$ is smaller. Using a common denominator of 12, $-\frac{1}{3}$ is $-\frac{4}{12}$ and $-\frac{1}{4}$ is $-\frac{3}{12}$, and $-\frac{4}{12}$ sits farther left on the number line. With negatives, the fraction with the bigger size is the smaller value.

#7 · Number

What is the easiest way to order a mixed list like -1.5, $\frac{2}{3}$, -2, 0.4?

Answer

Turn everything into decimals first, then line them up on a number line. Here $\frac{2}{3}$ is about 0.67, so the order from smallest to largest is -2, -1.5, 0.4, $\frac{2}{3}$.

#8 · Number

What does absolute value mean?

Answer

Absolute value is the size of a number without its sign, or its distance from zero on the number line. For example, the absolute value of -7 is 7, and the absolute value of 7 is also 7.

#9 · Number

What is the rule for adding two integers with the same sign?

Answer

Add their sizes and keep the shared sign. For example, $-3 + (-5) = -8$, because $3 + 5 = 8$ and both numbers are negative.

#10 · Number

What is the rule for adding two integers with different signs?

Answer

Subtract the smaller size from the larger size, then keep the sign of the number with the bigger size. For example, $-7 + 4 = -3$, because $7 - 4 = 3$ and the bigger size (7) was negative.

#11 · Number

How do you handle any integer subtraction problem?

Answer

Change it into adding the opposite: $a - b$ becomes $a + (-b)$. For example, $5 - (-3)$ becomes $5 + 3 = 8$. This one rewrite turns every subtraction into an addition question you already know how to solve.

#12 · Number

What is $-6 - (-9)$?

Answer

The answer is 3. Rewrite it as adding the opposite: $-6 + 9$. The signs are different, so subtract 6 from 9 to get 3, and keep the sign of the bigger size, which is positive.

#13 · Number

What mistake do students make when subtracting a negative, like $8 - (-5)$?

Answer

They forget to flip the subtraction into addition and wrongly write $8 - 5 = 3$. Subtracting a negative is the same as adding a positive, so $8 - (-5) = 8 + 5 = 13$. Always rewrite as adding the opposite first.

#14 · Number

What is $2 - 8$?

Answer

The answer is -6. Rewrite as $2 + (-8)$. The signs are different, so subtract 2 from 8 to get 6, and keep the negative sign because 8 was the bigger size.

#15 · Number

What is the sign rule for multiplying two negatives?

Answer

A negative times a negative gives a positive. Same signs always give a positive answer when multiplying or dividing. For example, $(-4) \times (-5) = 20$.

#16 · Number

What is the sign rule for multiplying or dividing numbers with different signs?

Answer

Different signs give a negative answer. For example, $(-4) \times 5 = -20$, and 30 divided by $(-6) = -5$. The rule is the same for both multiplication and division.

#17 · Number

Quick trick: what sign is $(-2) \times (-3) \times (-1) \times 4$?

Answer

Count the negative signs. There are three, an odd number, so the answer is negative. Multiplying the sizes gives $2 \times 3 \times 1 \times 4 = 24$, so the answer is -24.

#18 · Number

What must you do before adding or subtracting two fractions?

Answer

Give them a common denominator, meaning rewrite both fractions with the same bottom number. Only then can you add or subtract the numerators. The denominator stays the same in the answer.

#19 · Number

What is $\frac{1}{2} + \frac{2}{3}$?

Answer

The answer is $\frac{7}{6}$, or 1 and $\frac{1}{6}$. The common denominator of 2 and 3 is 6, so $\frac{1}{2}$ becomes $\frac{3}{6}$ and $\frac{2}{3}$ becomes $\frac{4}{6}$. Adding the numerators gives $\frac{3}{6} + \frac{4}{6} = \frac{7}{6}$.

#20 · Number

What mistake do students make when adding fractions like $\frac{1}{2} + \frac{1}{3}$?

Answer

They add straight across the top and bottom, wrongly getting $\frac{2}{5}$. The correct method uses a common denominator of 6: $\frac{1}{2}$ becomes $\frac{3}{6}$, $\frac{1}{3}$ becomes $\frac{2}{6}$, and $\frac{3}{6} + \frac{2}{6} = \frac{5}{6}$.

#21 · Number

How do you multiply two fractions?

Answer

Multiply the numerators together and the denominators together, then simplify. For example, $\frac{2}{3} \times \frac{3}{5} = \frac{6}{15}$, which simplifies to $\frac{2}{5}$ by dividing top and bottom by 3. No common denominator is needed for multiplying.

#22 · Number

How do you divide one fraction by another?

Answer

Flip the second fraction (take its reciprocal) and multiply. For example, $\frac{3}{4}$ divided by $\frac{2}{5}$ becomes $\frac{3}{4} \times \frac{5}{2} = \frac{15}{8}$, which is 1 and $\frac{7}{8}$.

#23 · Number

What is a reciprocal?

Answer

The reciprocal of a fraction is what you get when you swap its numerator and denominator. The reciprocal of $\frac{2}{5}$ is $\frac{5}{2}$. You use it when dividing fractions: flip the second fraction, then multiply.

#24 · Number

What is $-\frac{3}{4} - \frac{1}{2}$?

Answer

The answer is $-\frac{5}{4}$, or -1 and $\frac{1}{4}$. Rewrite as $-\frac{3}{4} + (-\frac{1}{2})$, give them the common denominator 4 so $-\frac{1}{2}$ becomes $-\frac{2}{4}$, then add: $-\frac{3}{4} + (-\frac{2}{4}) = -\frac{5}{4}$.

#25 · Number

What is the rule for lining up decimals when adding or subtracting?

Answer

Line up the decimal points so ones sit under ones, tenths under tenths, and so on. For example, $4.75 + (-1.3)$ becomes $4.75 - 1.3 = 3.45$.

#26 · Number

What is 0.6×0.4 , and how do you place the decimal point?

Answer

The answer is 0.24. Multiply as if there were no decimals: $6 \times 4 = 24$. Then count the total decimal places in both numbers (one each, so two total) and move the point two spots from the right.

#27 · Number

What does BEDMAS stand for?

Answer

Brackets, Exponents, Division and Multiplication (left to right), Addition and Subtraction (left to right). It is the order you must follow when an expression has more than one operation.

#28 · Number

In BEDMAS, does multiplication always come before division?

Answer

No. Division and multiplication have equal priority and are done left to right in the order they appear. The same is true for addition and subtraction.

#29 · Number

What is 12 divided by 3×2 ?

Answer

The answer is 8. Work left to right: 12 divided by 3 is 4, then 4×2 is 8. A common wrong answer is 2, from doing 3×2 first, but division and multiplication share equal priority left to right.

#30 · Number

Solve: $3 + 2 \times (5 - 1)^2$ divided by 4.

Answer

The answer is 11. Brackets first: $5 - 1 = 4$. Exponent next: $4^2 = 16$. Then multiply and divide left to right: $2 \times 16 = 32$, and 32 divided by 4 = 8. Finally add: $3 + 8 = 11$.

#31 · Number

What are nested brackets, and how do you handle them?

Answer

Nested brackets are brackets sitting inside other brackets. Always solve the innermost brackets first, then work your way outward. For example, in $-2 \times [(-3 + 7) - 2^2]$, solve $-3 + 7 = 4$ and $2^2 = 4$ first, giving $-2 \times 0 = 0$.

#32 · Number

Solve: $(1/2 + 1/4) \times (-8)$.

Answer

The answer is -6. Brackets first: $1/2 + 1/4 = 2/4 + 1/4 = 3/4$. Then multiply: $3/4 \times (-8) = -24/4 = -6$. The sign is negative because a positive times a negative is negative.

#33 · Number

In the power 5^3 , what are the base and the exponent, and what does it equal?

Answer

The base is 5 and the exponent is 3. The exponent tells you how many times to multiply the base by itself: $5 \times 5 \times 5 = 125$.

#34 · Number

What is the product rule for exponents?

Answer

When multiplying two powers with the same base, keep the base and add the exponents. For example, $2^3 \times 2^4 = 2^7 = 128$. It only works when the bases are the same.

#35 · Number

What is the quotient rule for exponents?

Answer

When dividing two powers with the same base, keep the base and subtract the exponents. For example, 6^5 divided by $6^2 = 6^3 = 216$.

#36 · Number

What is the power of a power rule?

Answer

When a power is raised to another exponent, keep the base and multiply the exponents. For example, $(3^2)^4 = 3^8$. A quick check: $(3^2)^2 = 9^2 = 81$, and 3^4 also equals 81.

#37 · Number

What does any nonzero number to the exponent zero equal?

Answer

It equals 1. For example, $9^0 = 1$ and $(-15)^0 = 1$. This keeps the quotient rule consistent, since 4^3 divided by 4^3 must be 1, and subtracting exponents gives 4^0 .

#38 · Number

What does a negative exponent mean?

Answer

It means take the reciprocal of the base, and the exponent becomes positive. For example, $2^{-3} = 1/(2^3) = 1/8$, and $5^{-2} = 1/25$. A negative exponent does not make the answer negative.

#39 · Number

What is the difference between $(-2)^3$ and 2^{-3} ?

Answer

$(-2)^3$ has a negative base: $(-2) \times (-2) \times (-2) = -8$. But 2^{-3} has a negative exponent, which means the reciprocal: $1/8$, a positive fraction. The position of the negative sign completely changes the meaning.

#40 · Number

Why is $(-3)^2$ not the same as -3^2 ?

Answer

$(-3)^2$ means $(-3) \times (-3) = 9$, because the brackets put the negative sign inside the power. But -3^2 means $-(3^2) = -9$, because without brackets the exponent only applies to the 3. Always check where the negative sign sits.

#41 · Number

What common exponent mistake happens with $2^3 \times 3^2$?

Answer

Students wrongly add the exponents and write 6^5 . The product rule only works when the bases are the same, and here they are different. Calculate each power separately: $2^3 = 8$ and $3^2 = 9$, so the answer is $8 \times 9 = 72$.

#42 · Number

What is estimation used for on a test?

Answer

Estimation gives a quick, close answer that checks whether your exact answer makes sense. If your exact answer is wildly different from your estimate, look for a sign error or a misplaced decimal point. Use it as a check, not a replacement for the real calculation.

#43 · Number

What is the rounding-to-friendly-numbers estimation strategy?

Answer

Round each number to a nearby easy number before calculating. For example, to estimate $(6.8 + 3.2)$ divided by 2, round to 7 and 3: $7 + 3 = 10$, then 10 divided by $2 = 5$. The exact answer is also 5, so the estimate confirms it.

#44 · Number

What are compatible numbers in estimation?

Answer

Compatible numbers are nearby numbers that divide or multiply evenly. To estimate 79 divided by 8, use 80 divided by $8 = 10$, so the real answer is a little under 10. The exact answer is 9.875.

#45 · Number

How do you estimate a sum of fractions like $\frac{7}{8} + \frac{1}{9}$?

Answer

Compare each fraction to 0, $\frac{1}{2}$, or 1. Here $\frac{7}{8}$ is close to 1 and $\frac{1}{9}$ is close to 0, so the sum should land near 1. The exact answer, $\frac{71}{72}$, is just under 1, matching the estimate.

#46 · Number

What is a quick mental shortcut for finding 10% of any number?

Answer

Move the decimal point one place to the left. For example, 10% of 350 is 35. From there, 5% is half of the 10% value (17.5) and 1% is one tenth of it (3.5), so you can build almost any percent quickly.

#47 · Number

The temperature is -12 degrees and rises 8 degrees. What is it now?

Answer

It is -4 degrees. Set it up as $-12 + 8$. The signs are different, so subtract 8 from 12 to get 4, and keep the negative sign because 12 was the bigger size.

#48 · Number

What is a ratio?

Answer

A ratio compares two or more quantities, showing how much of one thing there is compared to another. It can be written with a colon, like 12:18, or as a fraction, like $12/18$. The order of the numbers matters.

#49 · Number

What is the difference between a part-to-part and a part-to-whole ratio?

Answer

A part-to-part ratio compares one part of a group to another part, like boys to girls, 12:18. A part-to-whole ratio compares one part to the entire group, like boys to the whole class, 12:30.

#50 · Number

How do you simplify a ratio like 12:18?

Answer

Divide both numbers by their greatest common factor. The greatest common factor of 12 and 18 is 6, so the simplified ratio is 2:3. The smaller ratio still represents the exact same relationship.

#51 · Number

What are equivalent ratios?

Answer

Equivalent ratios show the same comparison with different numbers. You make one by multiplying or dividing both parts by the same value. For example, 2:3 is equivalent to 4:6, 6:9, and 20:30.

#52 · Number

**A recipe uses flour to sugar in a 3:2 ratio.
How much sugar goes with 9 cups of flour?**

Answer

6 cups of sugar. The flour went from 3 to 9, a scale factor of 3, so multiply the sugar by the same factor:
 $2 \times 3 = 6$.

#53 · Number

Paint mixes blue to white at 5:3. You have 40 L of blue. How much white?

Answer

24 litres. Find the scale factor: 40 divided by 5 is 8. Multiply the white part by the same factor: $3 \times 8 = 24$.

#54 · Number

A basket has apples to oranges in a 2:3 ratio and 20 fruits total. How many apples?

Answer

8 apples. The ratio has $2 + 3 = 5$ parts, and 20 divided by 5 means each part is 4 fruits. Apples get 2 parts: $2 \times 4 = 8$, and oranges get $3 \times 4 = 12$, which checks out since $8 + 12 = 20$.

#55 · Number

Why does order matter in a ratio?

Answer

A ratio of boys to girls, 12:18, is not the same as girls to boys, 18:12, even though the numbers are identical. Whatever the question names first must stay first all the way through your solution. Switching partway loses marks even with correct arithmetic.

#56 · Number

What is a rate?

Answer

A rate is a special ratio comparing two quantities with different units. Speed compares distance to time, like kilometres per hour, and price compares cost to quantity, like dollars per kilogram.

#57 · Number

What is a unit rate?

Answer

A unit rate is a rate where the second quantity is 1, like 60 km per hour or \$0.50 per apple. Unit rates make it easy to compare speeds or prices, because everything is scaled to per one.

#58 · Number

A car travels 240 km in 4 hours. What is its unit rate?

Answer

60 kilometres per hour. Divide the distance by the time: 240 divided by 4 equals 60.

#59 · Number

A bag of 8 apples costs \$4.00. What is the unit price?

Answer

\$0.50 per apple. Divide the cost by the quantity:
\$4.00 divided by 8 equals \$0.50.

#60 · Number

How do you decide which product is the better buy?

Answer

Convert both prices to the same unit price, like dollars per gram, then compare those directly. For example, a 500 g box at \$5.00 is \$0.01 per gram, while a 750 g box at \$6.75 is \$0.009 per gram, so the bigger box is the better deal.

#61 · Number

What trap do students fall into when comparing prices at the store?

Answer

They compare total prices instead of unit prices. A bigger box can cost more in total but still be cheaper per gram because it holds much more product. Always convert to the same unit before deciding.

#62 · Number

What trap happens with the direction of a unit rate?

Answer

Dividing the wrong way flips the rate. If a question asks for kilometres per hour, divide distance by time, not time by distance. Always check that your final units match what the question asks for.

#63 · Number

What is a proportion?

Answer

A proportion is a statement that two ratios or rates are equal, written as two fractions set equal, like $a/b = c/d$. Solving a proportion means finding the one missing value when the other three are known.

#64 · Number

How does cross-multiplication work?

Answer

If $a/b = c/d$, then a times d equals b times c . Multiply diagonally across the equals sign, set the two products equal, and solve for the missing value.

#65 · Number

Solve for x: $\frac{3}{4} = \frac{x}{20}$.

Answer

$x = 15$. Cross-multiply: $3 \times 20 = 4x$, so $60 = 4x$, and dividing both sides by 4 gives 15. Check: $15/20$ simplifies to $3/4$.

#66 · Number

What does it mean for two quantities to be in direct proportion?

Answer

They increase or decrease together at the same rate: double one and the other doubles too. A quick test is dividing one by the other, which always gives the same constant value, like a fixed hourly wage.

#67 · Number

Is a taxi fare with a flat starting fee a direct proportion?

Answer

No. Doubling the distance does not double the total fare, because the flat fee stays the same either way. Only relationships where doubling one quantity doubles the other are direct proportions.

#68 · Number

What does the word percent mean?

Answer

Percent means per hundred. A percentage is a ratio comparing a number to 100, so 45% means 45 out of every 100.

#69 · Number

How do you convert a percentage to a decimal?

Answer

Divide by 100, which moves the decimal point two places left. For example, 45% becomes 0.45 and 7% becomes 0.07.

#70 · Number

How do you convert a decimal to a percentage?

Answer

Multiply by 100, which moves the decimal point two places right. For example, 0.6 becomes 60% and 0.125 becomes 12.5%.

#71 · Number

Convert $\frac{3}{8}$ to a percentage.

Answer

37.5%. Divide the numerator by the denominator: 3 divided by 8 equals 0.375. Then multiply by 100 to get 37.5%.

#72 · Number

Convert 60% to a fraction in lowest terms.

Answer

$\frac{3}{5}$. Write it over 100 as $\frac{60}{100}$, then divide top and bottom by their greatest common factor, 20.

#73 · Number

What is 25% of 160?

Answer

40. Convert the percent to a decimal, 0.25, then multiply: $0.25 \times 160 = 40$.

#74 · Number

What is 8% of 350?

Answer

28. Convert 8% to the decimal 0.08, then multiply: $0.08 \times 350 = 28$.

#75 · Number

What is 15% of 80?

Answer

12. Convert 15% to 0.15 and multiply: $0.15 \times 80 = 12$. Or use the shortcut: 10% of 80 is 8, and 5% is half of that (4), so $8 + 4 = 12$.

#76 · Number

A \$40 shirt increases in price by 15%. What is the new price?

Answer

\$46. Find the increase: $0.15 \times 40 = \$6$. Then add it to the original: $\$40 + \$6 = \$46$.

#77 · Number

An \$80 jacket is 30% off. What is the sale price?

Answer

\$56. Find the decrease: $0.30 \times 80 = \$24$. Then subtract from the original: $\$80 - \$24 = \$56$.

#78 · Number

What is the formula for percent change?

Answer

Percent change equals the amount of change divided by the original amount, times 100. For example, a population growing from 200 to 250 changed by 50, and 50 divided by 200 is 0.25, so the increase is 25%.

#79 · Number

What trap mixes up 'percent of' with 'percent increase'?

Answer

Finding 20% of 80 gives 16, but increasing 80 by 20% gives 96, the original plus the increase. These are two completely different calculations, so read the question carefully to see which one it asks for.

#80 · Number

After a 20% increase, a bike costs \$96. What was the original price?

Answer

\$80. Set up the equation: $\text{original} \times 1.20 = 96$, then divide both sides by 1.20. Check: 20% of \$80 is \$16, and $\$80 + \$16 = \$96$.

#81 · Number

**After a 25% discount, a jacket costs \$63.
What was the original price?**

Answer

\$84. Set up: $\text{original} \times 0.75 = 63$, since 100% minus 25% leaves 75%. Divide both sides by 0.75 to get \$84. Check: 25% of \$84 is \$21, and $\$84 - \$21 = \$63$.

#82 · Number

**What mistake do students make when
finding an original price before a discount?**

Answer

They take the percent of the new price and add it back, which gives the wrong answer. The percent was calculated from the original price, not the new one. Always write $\text{original} \times (1 \text{ plus or minus the percent}) = \text{new amount}$, then solve for the original.

#83 · Number

What is Ontario's HST rate, and how do you add it to a price?

Answer

The HST is 13%. Multiply the purchase price by 1.13 to get the total with tax. For example, a \$50 item costs $50 \times 1.13 = \$56.50$ in total.

#84 · Number

A \$120 item is 20% off, then 13% HST is added. What is the final price?

Answer

\$108.48. Apply the discount first: $120 \times 0.80 = \$96$. Then apply the tax to the discounted price: $96 \times 1.13 = \$108.48$. Each step uses the current price, not the original.

#85 · Number

What does a scale of 1:100 on a diagram mean?

Answer

One unit on the diagram equals 100 of the same unit in real life. So 1 cm drawn on the page represents 100 cm of real distance.

#86 · Number

A diagram uses a 1:50 scale. A wall is 6 cm on the diagram. Real length?

Answer

300 cm, which is 3 metres. Multiply the drawn length by the scale: $6 \times 50 = 300$. You can also set up the proportion $1/50 = 6/x$ and cross-multiply.

#87 · Number

A map scale says 2 cm = 15 km. Two cities are 7 cm apart on the map. Real distance?

Answer

52.5 km. Set up the proportion $2/15 = 7/x$ and cross-multiply: $2x = 15 \times 7 = 105$. Divide both sides by 2 to get 52.5.

#88 · Number

A recipe serves 4 people and uses 2 cups of rice. How much rice for 10 people?

Answer

5 cups. Find the scale factor: 10 divided by 4 is 2.5. Multiply the rice by that factor: $2 \times 2.5 = 5$. Every other ingredient gets multiplied by the same 2.5.

#89 · Number

A recipe calls for $\frac{3}{4}$ cup of flour and you make 1.5 times the recipe. How much flour?

Answer

1 and $\frac{1}{8}$ cups. Since 1.5 equals $\frac{3}{2}$, multiply: $\frac{3}{4} \times \frac{3}{2} = \frac{9}{8}$, which is 1 and $\frac{1}{8}$.

#90 · Number

A cleaner mixes 1 part concentrate to 4 parts water. How do you make 15 L of solution?

Answer

Use 3 L of concentrate and 12 L of water. The ratio has $1 + 4 = 5$ parts, and 15 divided by 5 means each part is 3 litres. Concentrate gets 1 part (3 L) and water gets 4 parts (12 L).

#91 · Algebra

What is a variable?

Answer

A variable is a letter that stands in for a number we do not know yet, or a number that can change. The most common letters are x , y , and n , but any letter works. When you see x , just think 'some number.'

#92 · Algebra

What is the difference between an expression and an equation?

Answer

An expression is a math phrase with numbers, variables, and operations but no equals sign, like $3x + 5$. An equation has an equals sign, like $3x + 5 = 20$. You simplify expressions, but you solve equations.

#93 · Algebra

Translate into algebra: five more than a number.

Answer

Write it as $n + 5$. 'More than' signals addition, so you add 5 to the unknown number n .

#94 · Algebra

Translate into algebra: seven less than twice a number.

Answer

Write it as $2n - 7$. 'Twice a number' means $2n$, and 'seven less than' means you subtract 7 from it. Watch the order: it is $2n - 7$, not $7 - 2n$.

#95 · Algebra

Translate: a number decreased by 3, then multiplied by 2.

Answer

Write it as $2(n - 3)$. The word 'then' tells you the subtraction happens first, so it needs brackets. Writing $2n - 3$ would be wrong because that only multiplies the n by 2, not the whole quantity.

#96 · Algebra

Which words in a problem usually signal addition?

Answer

Words like sum, total, more than, and increased by usually mean addition. For example, 'the sum of a number and 4' becomes $n + 4$.

#97 · Algebra

Which words in a problem usually signal multiplication?

Answer

Words like product, times, and of usually mean multiplication. For example, 'the product of three and a number' becomes $3n$.

#98 · Algebra

Say the expression $4x - 9$ in words.

Answer

One correct way is 'nine less than four times a number.' There is often more than one correct wording, as long as the math meaning stays the same.

#99 · Algebra

Trap: what does 'less than' do to the order of a subtraction?

Answer

'Less than' flips the order. 'Seven less than twice a number' means $2n - 7$, because the 7 is taken away from $2n$. Writing $7 - 2n$ reverses the meaning and gives a wrong answer.

#100 · Algebra

Translate: take a number, double it, then add seven.

Answer

Write it as $2n + 7$. Doubling the number gives $2n$, and adding seven gives $2n + 7$. Variables let us shrink a whole sentence into a short expression.

#101 · Algebra

What is a term in an algebraic expression?

Answer

A term is a single piece of an expression, separated from other pieces by a plus or minus sign. In $5x^2 + 3x - 7$, the terms are $5x^2$, $3x$, and -7 . The sign right in front of a term belongs to that term.

#102 · Algebra

What is a coefficient?

Answer

A coefficient is the number multiplied in front of a variable. In the term $5x^2$, the coefficient is 5. In $3x$, it is 3.

#103 · Algebra

What is the coefficient of the term x ?

Answer

The coefficient is 1, because $1x$ is exactly the same as x . When no number is written in front of a variable, an invisible 1 is understood to be there.

#104 · Algebra

What is a constant term?

Answer

A constant is a term with no variable at all, just a number by itself. In $5x^2 + 3x - 7$, the constant is -7. Constants never change value no matter what x is.

#105 · Algebra

What are like terms?

Answer

Like terms have the exact same variable part, including the exact same exponent. For example, $4x$ and $9x$ are like terms. Only like terms can be added or subtracted together.

#106 · Algebra

Are $4x$ and $4x^2$ like terms?

Answer

No. Even though the coefficients look the same, one term has x^1 and the other has x^2 . The variable parts are different, so they cannot be combined.

#107 · Algebra

Are $3xy$ and $8xy$ like terms?

Answer

Yes. Both terms have exactly the same variable part, xy , so they are like terms. They can be combined: $3xy + 8xy = 11xy$.

#108 · Algebra

Are $3xy$ and $3x$ like terms?

Answer

No. One term has both x and y while the other only has x . To be like terms, every letter and every exponent in the variable part must match exactly.

#109 · Algebra

How many terms are in $5x^2 + 3x - 7$?

Answer

Three terms: $5x^2$, $3x$, and -7 . Terms are the pieces separated by plus or minus signs, and each sign belongs to the term right after it.

#110 · Algebra

What is the coefficient of the term $-7x$?

Answer

The coefficient is -7 . The minus sign in front of a term belongs to that term, so the sign is part of the coefficient.

#111 · Algebra

What does it mean to simplify an expression?

Answer

Simplifying means rewriting an expression in a shorter, cleaner form without changing its value. The main tool is collecting like terms: add or subtract the coefficients of terms that are alike, and leave unlike terms alone.

#112 · Algebra

Simplify: $6x + 3x - 2x$

Answer

All three are like terms, so combine the coefficients: $6 + 3 - 2 = 7$. The answer is $7x$.

#113 · Algebra

Simplify: $4x + 7 - 2x + 5$

Answer

Group the x terms and the constants: $(4x - 2x) + (7 + 5)$. That gives $2x + 12$. You cannot combine $2x$ and 12 because they are not like terms.

#114 · Algebra

Simplify: $5x + 3y - 2x + 6y$

Answer

Group the x terms and the y terms separately: $(5x - 2x) + (3y + 6y)$. The answer is $3x + 9y$. The x and y terms cannot be combined into one because they are different variables.

#115 · Algebra

Simplify: $8a - 3b + 2a + b - 5$

Answer

Group the a terms, b terms, and the constant: $(8a + 2a) + (-3b + b) + (-5)$. That gives $10a - 2b - 5$. Remember that b by itself has a coefficient of 1.

#116 · Algebra

Trap: can $4x + 4x^2$ be simplified further?

Answer

No. The terms have different exponents, so they are not like terms and must stay separate. Adding them to get $8x$ or $8x^2$ is one of the most common mistakes in this unit.

#117 · Algebra

Simplify: $9m - 4m + m$

Answer

All three are like terms. Combine the coefficients: $9 - 4 + 1 = 6$. The answer is $6m$.

#118 · Algebra

Simplify: $2x + 3 + 5x - 8$

Answer

Group the x terms and the constants: $(2x + 5x) + (3 - 8)$. That gives $7x - 5$.

#119 · Algebra

State the distributive property.

Answer

The distributive property says $a(b + c) = ab + ac$. In plain words, multiply the outside term by every single term inside the brackets, one at a time.

#120 · Algebra

Expand: $3(2x + 5)$

Answer

Multiply 3 by each term inside. 3 times $2x$ is $6x$, and 3 times 5 is 15. The answer is $6x + 15$.

#121 · Algebra

Expand: $-2(x - 4)$

Answer

Multiply -2 by each term. -2 times x is $-2x$, and -2 times -4 is +8, because a negative times a negative is positive. The answer is $-2x + 8$, not $-2x - 8$.

#122 · Algebra

Expand: $4(3x^2 - 2x + 5)$

Answer

Multiply 4 by all three terms: 4 times $3x^2$ is $12x^2$, 4 times $-2x$ is $-8x$, and 4 times 5 is 20. The answer is $12x^2 - 8x + 20$. Missing the last term is an easy way to lose marks.

#123 · Algebra

Trap: what happens to signs when you distribute a negative number?

Answer

Every term inside the brackets flips sign, not just the first one. For example, $-2(x - 4)$ becomes $-2x + 8$. Sign errors while distributing a negative are the single most common mistake in this unit.

#124 · Algebra

Expand: $5(x - 3)$

Answer

Multiply 5 by each term: 5 times x is $5x$, and 5 times -3 is -15 . The answer is $5x - 15$.

#125 · Algebra

Expand: $-(2x - 7)$

Answer

A minus sign in front of brackets acts like multiplying by -1 , so every term flips sign. The answer is $-2x + 7$.

#126 · Algebra

**Why does the distributive property work?
(Think of a rectangle.)**

Answer

Picture a rectangle with width a and a length split into two pieces, b and c . The whole area is $a(b + c)$, but you could also add the two smaller areas, $ab + ac$. Both ways give the same total area, which is exactly what the rule says.

#127 · Algebra

Simplify: $2(3x + 1) + 4$

Answer

First expand: 2 times $3x$ is $6x$, and 2 times 1 is 2, giving $6x + 2 + 4$. Then collect the constants: $6x + 6$.

#128 · Algebra

What are a monomial, a binomial, and a trinomial?

Answer

They are polynomials named by their number of terms. A monomial has one term, like $3x$. A binomial has two terms, like $x + 5$. A trinomial has three terms, like $x^2 + 2x - 7$.

#129 · Algebra

What is a polynomial?

Answer

A polynomial is an algebraic expression made of one or more terms added or subtracted together, like $3x^2 + 2x - 5$. Each piece separated by a plus or minus sign is one term.

#130 · Algebra

Add: $(3x^2 + 2x - 5) + (x^2 - 4x + 7)$

Answer

Drop the brackets, since adding does not change any signs, then group like terms: $(3x^2 + x^2) + (2x - 4x) + (-5 + 7)$. The answer is $4x^2 - 2x + 2$.

#131 · Algebra

What is the rule for adding two polynomials?

Answer

Remove the brackets and collect like terms as usual. Adding brackets does not change any signs, so no terms flip. Just group the matching variable parts and combine their coefficients.

#132 · Algebra

What is the rule for subtracting one polynomial from another?

Answer

Distribute the negative sign to every term in the second bracket first, flipping every sign inside it. Then collect like terms as usual. This flip step is where most students lose marks.

#133 · Algebra

Subtract: $(5x^2 - 3x + 1) - (2x^2 - x - 4)$

Answer

Flip every sign in the second bracket: $-2x^2 + x + 4$.
Now combine: $(5x^2 - 2x^2) + (-3x + x) + (1 + 4)$. The answer is $3x^2 - 2x + 5$.

#134 · Algebra

Trap: when subtracting polynomials, which terms flip sign?

Answer

Every single term in the second bracket flips, not just the first one. Flipping only the first term is one of the most common ways to lose full marks on an otherwise correct answer.

#135 · Algebra

Subtract: $(4x + 9) - (x - 2)$

Answer

Flip both signs in the second bracket: $-x + 2$. Then combine: $(4x - x) + (9 + 2)$. The answer is $3x + 11$.

#136 · Algebra

Add: $(2x - 3) + (5x + 8)$

Answer

Drop the brackets and group like terms: $(2x + 5x) + (-3 + 8)$. The answer is $7x + 5$.

#137 · Algebra

How do you multiply a monomial by a polynomial?

Answer

Use the distributive property: multiply the single term by every term inside the brackets. Multiply the coefficients, and use the product rule to add exponents on matching variables.

#138 · Algebra

Multiply: $4x(3x^2 - 2x + 5)$

Answer

$4x$ times $3x^2$ is $12x^3$ (multiply 4 and 3, add exponents $1 + 2$). $4x$ times $-2x$ is $-8x^2$, and $4x$ times 5 is $20x$. The answer is $12x^3 - 8x^2 + 20x$.

#139 · Algebra

Multiply: $-3x^2(2x - 5)$

Answer

$-3x^2$ times $2x$ is $-6x^3$, and $-3x^2$ times -5 is $+15x^2$, since a negative times a negative is positive. The answer is $-6x^3 + 15x^2$.

#140 · Algebra

Multiply: $2x(x + 6)$

Answer

$2x$ times x is $2x^2$, and $2x$ times 6 is $12x$. The answer is $2x^2 + 12x$.

#141 · Algebra

What is the rule for multiplying two binomials?

Answer

Every term in the first bracket must be multiplied by every term in the second bracket, giving four products in total. Then collect like terms at the end, usually the two middle terms.

#142 · Algebra

Expand: $(x + 3)(x + 5)$

Answer

Multiply all four pairs: x times x is x^2 , x times 5 is $5x$, 3 times x is $3x$, and 3 times 5 is 15 . Collect the middle terms: $5x + 3x = 8x$. The answer is $x^2 + 8x + 15$.

#143 · Algebra

Expand: $(2x - 1)(x + 4)$

Answer

The four products are $2x^2$, $8x$, $-x$, and -4 . Collect the middle terms: $8x - x = 7x$. The answer is $2x^2 + 7x - 4$.

#144 · Algebra

Expand: $(x + 2)(x + 7)$

Answer

The four products are x^2 , $7x$, $2x$, and 14 . Collect the middle terms: $7x + 2x = 9x$. The answer is $x^2 + 9x + 14$.

#145 · Algebra

Expand: $(x - 3)(x + 3)$

Answer

The four products are x^2 , $3x$, $-3x$, and -9 . The middle terms cancel out: $3x - 3x = 0$. The answer is $x^2 - 9$.

#146 · Algebra

How does an area model help multiply two binomials?

Answer

Draw a rectangle split into a 2 by 2 grid, with the two binomials along the top and side. Each small rectangle is one of the four multiplications. Adding all four areas gives the expanded answer.

#147 · Algebra

Expand: $(x + 4)(x - 2)$

Answer

The four products are x^2 , $-2x$, $4x$, and -8 . Collect the middle terms: $-2x + 4x = 2x$. The answer is $x^2 + 2x - 8$.

#148 · Algebra

Exponent product rule: what happens when you multiply powers with the same base?

Answer

Add the exponents: x^m times x^n equals x^{m+n} . For example, x^3 times x^4 equals x^7 . This only works when the bases are the same.

#149 · Algebra

Simplify: x^3 times x^4

Answer

Same base, so add the exponents: $3 + 4 = 7$. The answer is x^7 .

#150 · Algebra

Simplify: $(2x^3)(3x^2)$

Answer

Multiply the coefficients first: 2 times 3 is 6. Then add the exponents: $3 + 2 = 5$. The answer is $6x^5$.

#151 · Algebra

Exponent quotient rule: what happens when you divide powers with the same base?

Answer

Subtract the exponents: x^m divided by x^n equals x^{m-n} . For example, x^8 divided by x^3 equals x^5 .

#152 · Algebra

Simplify: x^8 divided by x^3

Answer

Same base, so subtract the exponents: $8 - 3 = 5$.
The answer is x^5 .

#153 · Algebra

Simplify: $12x^7$ divided by $4x^3$

Answer

Divide the coefficients: 12 divided by 4 is 3. Subtract the exponents: $7 - 3 = 4$. The answer is $3x^4$.

#154 · Algebra

Power of a power rule: what happens in $(x^m)^n$?

Answer

Multiply the exponents: $(x^m)^n$ equals x^{mn} . For example, $(x^3)^2$ equals x^6 . It makes sense because $(x^3)^2$ means x^3 times x^3 , and the product rule gives x^6 .

#155 · Algebra

Simplify: $(x^3)^2$

Answer

A power raised to a power means multiply the exponents: 3 times 2 is 6. The answer is x^6 .

#156 · Algebra

Trap: when do you ADD exponents and when do you MULTIPLY them?

Answer

Add when multiplying same-base powers: x^3 times $x^4 = x^7$. Multiply when a power is raised to another power: $(x^3)^2 = x^6$. These are two different rules for two different situations, so read the expression carefully.

#157 · Algebra

Can x^3 and y^2 be combined with an exponent rule?

Answer

No. The exponent rules only work when the bases are the same, and x and y are different bases. The most you can do is write them multiplied together as x^3y^2 .

#158 · Algebra

Simplify: $(5x^2)(2x^4)$

Answer

Multiply the coefficients: 5 times 2 is 10. Add the exponents: $2 + 4 = 6$. The answer is $10x^6$.

#159 · Algebra

What is factoring?

Answer

Factoring is the reverse of distributing. Instead of multiplying a term across brackets, you pull a common factor out and rewrite the polynomial as that factor times a bracket.

#160 · Algebra

How do you find the greatest common factor (GCF) of a polynomial?

Answer

Find the biggest number that divides evenly into all the coefficients. Then find the smallest power of each variable that appears in every term. Multiply those together to get the GCF.

#161 · Algebra

Factor: $6x + 9$

Answer

The biggest number dividing both 6 and 9 is 3, and there is no variable in both terms, so the GCF is 3. Divide each term by 3 to get $2x$ and 3. The factored form is $3(2x + 3)$.

#162 · Algebra

Factor: $8x^2 - 12x$

Answer

The biggest number dividing 8 and 12 is 4, and both terms have at least one x , so the GCF is $4x$. Dividing each term by $4x$ gives $2x$ and -3 . The factored form is $4x(2x - 3)$.

#163 · Algebra

Factor: $10x^3 + 15x^2 - 5x$

Answer

The GCF of 10, 15, and 5 is 5, and every term has at least one x, so the GCF is 5x. Dividing each term by 5x gives $2x^2$, 3x, and -1. The factored form is $5x(2x^2 + 3x - 1)$.

#164 · Algebra

How do you check a factoring answer?

Answer

Distribute it back out and see if you get the original expression. For example, $3(2x + 3)$ expands to $6x + 9$, so the factoring was correct. This check takes seconds and catches most errors.

#165 · Algebra

Factor: $5x + 20$

Answer

The biggest number dividing 5 and 20 is 5, so the GCF is 5. Dividing each term by 5 gives x and 4. The factored form is $5(x + 4)$.

#166 · Algebra

Factor: $7x^2 + 14x$

Answer

The GCF of the coefficients is 7, and both terms have at least one x , so the GCF is $7x$. Dividing each term by $7x$ gives x and 2. The factored form is $7x(x + 2)$.

#167 · Algebra

What is substitution, and what order of operations do you use?

Answer

Substitution means replacing a variable with a specific number and then simplifying. Always follow BEDMAS: Brackets, Exponents, Division and Multiplication, Addition and Subtraction.

#168 · Algebra

Evaluate $3x + 4$ when $x = 5$.

Answer

Replace x with 5: $3(5) + 4$. Multiply first to get $15 + 4$, then add. The answer is 19.

#169 · Algebra

Evaluate $3x^2 - 2x + 5$ when $x = -2$.

Answer

Substitute with brackets: $3(-2)^2 - 2(-2) + 5$. The exponent first: $(-2)^2 = 4$. Then multiply: $3(4) = 12$ and $-2(-2) = +4$. Adding gives $12 + 4 + 5 = 21$.

#170 · Algebra

Evaluate $2a - 3b$ when $a = 4$ and $b = -1$.

Answer

Substitute both letters: $2(4) - 3(-1)$. Multiplying gives $8 - (-3)$, which is the same as $8 + 3$. The answer is 11.

#171 · Algebra

Trap: why must you put substituted negative numbers in brackets?

Answer

Because -2^2 and $(-2)^2$ mean different things. Without brackets, only the 2 gets squared, giving -4, but $(-2)^2$ means -2 times -2, which is +4. Always bracket negatives to avoid sign errors.

#172 · Algebra

Evaluate $5x - 7$ when $x = 3$.

Answer

Substitute: $5(3) - 7$. Multiply first to get $15 - 7$. The answer is 8.

#173 · Algebra

Evaluate $x^2 + 2x$ when $x = -3$.

Answer

Substitute with brackets: $(-3)^2 + 2(-3)$. The exponent first: $(-3)^2 = 9$. Then $2(-3) = -6$. Adding gives $9 - 6 = 3$.

#174 · Algebra

What is a linear equation?

Answer

A linear equation is a math sentence saying two things are equal, where the variable is only raised to the power of 1. There is no x squared or x cubed anywhere. Solving it means finding the number that makes the sentence true.

#175 · Algebra

How is an equation like a balance scale?

Answer

Both sides must always stay equal, just like both pans must weigh the same. Whatever you do to one side (add, subtract, multiply, divide), you must do the exact same thing to the other side, or the equation breaks.

#176 · Algebra

What does it mean to isolate the variable?

Answer

It means getting x completely alone on one side of the equals sign. You do it by undoing operations in reverse order using opposite operations: addition undoes subtraction, and multiplication undoes division.

#177 · Algebra

Solve: $x + 7 = 15$

Answer

Subtract 7 from both sides to undo the addition: $x = 15 - 7 = 8$. Check: $8 + 7 = 15$, so $x = 8$ is correct.

#178 · Algebra

Solve: $4x = 20$

Answer

x is being multiplied by 4, so divide both sides by 4: $x = 20 / 4 = 5$. Check: $4(5) = 20$, so $x = 5$ is correct.

#179 · Algebra

Solve: $x - 9 = 4$

Answer

Add 9 to both sides to undo the subtraction: $x = 4 + 9 = 13$. Check: $13 - 9 = 4$, so $x = 13$ is correct.

#180 · Algebra

Solve: $x/5 = 6$

Answer

x is being divided by 5, so multiply both sides by 5:
 $x = 6 \text{ times } 5 = 30$. Check: $30 / 5 = 6$, so $x = 30$ is correct.

#181 · Algebra

In a two-step equation, which operation do you undo first?

Answer

Undo addition or subtraction first, then undo multiplication or division. This is the reverse of the order the equation was built in. For $3x + 5 = 20$, subtract 5 first, then divide by 3.

#182 · Algebra

Solve: $3x + 5 = 20$

Answer

Subtract 5 from both sides: $3x = 15$. Then divide both sides by 3: $x = 5$. Check: $3(5) + 5 = 20$, so $x = 5$ is correct.

#183 · Algebra

Solve: $x/4 - 3 = 5$

Answer

Add 3 to both sides: $x/4 = 8$. Then multiply both sides by 4: $x = 32$. Check: $32/4 - 3 = 8 - 3 = 5$, so $x = 32$ is correct.

#184 · Algebra

Solve: $2x - 7 = 11$

Answer

Add 7 to both sides: $2x = 18$. Divide both sides by 2: $x = 9$. Check: $2(9) - 7 = 18 - 7 = 11$, so $x = 9$ is correct.

#185 · Algebra

Solve: $5x + 2 = 17$

Answer

Subtract 2 from both sides: $5x = 15$. Divide both sides by 5: $x = 3$. Check: $5(3) + 2 = 17$, so $x = 3$ is correct.

#186 · Algebra

How do you check the solution to an equation?

Answer

Substitute your answer back into the original equation and see if both sides come out equal. It only takes a few seconds and catches almost every mistake before it costs you marks.

#187 · Algebra

Solve: $3(x + 2) = 2x + 10$

Answer

Expand the brackets first: $3x + 6 = 2x + 10$. Subtract $2x$ from both sides: $x + 6 = 10$. Subtract 6: $x = 4$. Check: $3(4) = 12$ and $2(4) + 10 = 18$, so both sides match.

#188 · Algebra

Solve: $(x + 1)/2 = 5$

Answer

Multiply both sides by 2 to clear the fraction: $x + 1 = 10$. Subtract 1: $x = 9$. Check: $(9 + 1)/2 = 10/2 = 5$, so $x = 9$ is correct.

#189 · Algebra

What is the first move when x appears on both sides of an equation?

Answer

Get every x term onto the same side. Pick the side with the smaller x term and subtract it from both sides so it disappears from that side. Then solve the leftover two-step equation as usual.

#190 · Algebra

Solve: $5x - 3 = 2x + 12$

Answer

Subtract $2x$ from both sides: $3x - 3 = 12$. Add 3: $3x = 15$. Divide by 3: $x = 5$. Check: $5(5) - 3 = 22$ and $2(5) + 12 = 22$, so both sides match.

#191 · Algebra

Solve: $7x = 4x + 9$

Answer

Subtract $4x$ from both sides: $3x = 9$. Divide by 3: $x = 3$. Check: $7(3) = 21$ and $4(3) + 9 = 21$, so $x = 3$ is correct.

#192 · Algebra

Solve: $6x + 1 = 2x + 13$

Answer

Subtract $2x$ from both sides: $4x + 1 = 13$. Subtract 1: $4x = 12$. Divide by 4: $x = 3$. Check: $6(3) + 1 = 19$ and $2(3) + 13 = 19$, so both sides match.

#193 · Algebra

Trap: when checking an answer, which version of the equation do you use?

Answer

Always plug your answer into the ORIGINAL equation, not a step from the middle of your work. If you made a mistake partway through, checking against a middle step would wrongly confirm the bad answer.

#194 · Algebra

Solve: $4(x - 1) = 2x + 6$

Answer

Expand first: $4x - 4 = 2x + 6$. Subtract $2x$: $2x - 4 = 6$. Add 4: $2x = 10$, so $x = 5$. Check: $4(5 - 1) = 16$ and $2(5) + 6 = 16$, so both sides match.

#195 · Algebra

What is a linear relation?

Answer

A relation is any rule connecting two sets of numbers, usually x and y . A linear relation is one whose graph forms a perfectly straight line. Not every relation is linear, so we test with first differences.

#196 · Algebra

What are first differences?

Answer

First differences are the amounts the y -values change by each time you move down one row in a table of values, as long as the x -values go up by the same amount each time. You find them by subtracting each y -value from the next one.

#197 · Algebra

How do first differences tell you if a relation is linear?

Answer

If the first differences are all the same (constant), the relation is linear and its graph is a straight line. If the differences change, the graph is curved and the relation is not linear.

#198 · Algebra

Table: x is 0,1,2,3 and y is 5,8,11,14. Linear?

Answer

Find the first differences: $8 - 5 = 3$, $11 - 8 = 3$, and $14 - 11 = 3$. Every difference is exactly 3, so they are constant. The relation is linear.

#199 · Algebra

A table's first differences are 3, then 5, then 8. Linear?

Answer

No. The first differences are not the same every time, so the graph would be curved instead of a straight line. Only constant first differences mean a linear relation.

#200 · Algebra

How are first differences connected to slope?

Answer

When the x-values in a table go up by 1 each time, the constant first difference IS the slope. It tells you how much y changes for every 1 unit increase in x.

#201 · Algebra

Trap: the x-values in a table jump by 2, not 1. How do you find slope?

Answer

You cannot use the change in y alone. Divide the change in y by the actual change in x for that row. For example, if y rises 6 while x rises 2, the slope is $6 / 2 = 3$.

#202 · Algebra

What does slope measure?

Answer

Slope measures how steep a line is and which way it tilts. In real life it is a rate of change, like cost per item or distance per hour. It is usually written with the letter m.

#203 · Algebra

What is the slope formula for two points?

Answer

For points (x_1, y_1) and (x_2, y_2) , the slope is $m = (y_2 - y_1) / (x_2 - x_1)$. It is remembered as rise over run: the change in y divided by the change in x.

#204 · Algebra

Find the slope between (1, 4) and (3, 10).

Answer

Use the formula: $m = (10 - 4) / (3 - 1) = 6 / 2 = 3$.
The slope is 3, meaning the line goes up 3 units for every 1 unit you move right.

#205 · Algebra

In 'rise over run,' what are rise and run?

Answer

Rise is the vertical change, meaning how far you move up or down (the change in y). Run is the horizontal change, meaning how far you move left or right (the change in x). Slope is rise divided by run.

#206 · Algebra

How do you find slope directly from a graph?

Answer

Pick any two clear points on the line. Count the rise (vertical squares between them) and the run (horizontal squares between them), then divide rise by run.

#207 · Algebra

What do positive and negative slopes look like?

Answer

A positive slope rises as you move right, like climbing a hill. A negative slope falls as you move right, like going down a ski hill.

#208 · Algebra

What are the slopes of horizontal and vertical lines?

Answer

A horizontal line has a slope of zero because it is flat. A vertical line has an undefined slope, because the run would be zero and you cannot divide by zero.

#209 · Algebra

Trap: does subtraction order matter in the slope formula?

Answer

Yes. Keep the same order on top and bottom: y_2 minus y_1 over x_2 minus x_1 . Mixing the order in only one part flips the sign of your slope and gives a wrong answer.

#210 · Algebra

Trap: what happens if you compute run over rise instead of rise over run?

Answer

You get the reciprocal of the real slope, which is a completely different number. Rise (the y change) always goes on top, and run (the x change) always goes on the bottom.

#211 · Algebra

Find the slope between (0, 4) and (2, 8).

Answer

Use the formula: $m = (8 - 4) / (2 - 0) = 4 / 2 = 2$. The slope is 2.

#212 · Algebra

What is the y-intercept?

Answer

The y-intercept is the point where a line crosses the y-axis. At that exact point, x always equals 0. In the equation $y = mx + b$, the y-intercept is the value b.

#213 · Algebra

What does the y-intercept usually mean in a real-world problem?

Answer

It is the starting value, the amount you begin with before anything changes. For example, a \$30 gym sign-up fee is the y-intercept because you pay it before any months have passed ($x = 0$).

#214 · Algebra

How do you spot the y-intercept in a table of values?

Answer

Find the row where $x = 0$. The y-value in that row is the y-intercept. In the table $x: 0, 1, 2, 3$ with $y: 5, 8, 11, 14$, the y-intercept is 5.

#215 · Algebra

In $y = mx + b$, what do m and b stand for?

Answer

m is the slope, the rate of change of the line. b is the y-intercept, the starting value where the line crosses the y-axis. Knowing both lets you write the full equation right away.

#216 · Algebra

Write the equation for the table x : 0,1,2,3 and y : 5,8,11,14.

Answer

The first differences are all 3, so the slope is $m = 3$. The y-value at $x = 0$ is 5, so $b = 5$. The equation is $y = 3x + 5$. Check with $x = 2$: $3(2) + 5 = 11$, which matches the table.

#217 · Algebra

A line crosses the y-axis at (0, 4) and passes through (2, 8). Equation?

Answer

The y-intercept gives $b = 4$. The slope is $m = (8 - 4) / (2 - 0) = 2$. So the equation is $y = 2x + 4$.

#218 · Algebra

A phone plan costs a \$20 base fee plus \$2 per GB. Equation and cost for 6 GB?

Answer

The base fee is the y-intercept ($b = 20$) and the cost per GB is the slope ($m = 2$), so $\text{Cost} = 2g + 20$. For 6 GB, $\text{Cost} = 2(6) + 20 = \32 .

#219 · Algebra

A taxi charges \$2 per kilometre. What does the slope of the cost graph mean?

Answer

The slope is 2, and it means the cost rises by \$2 for every extra kilometre travelled. Slope is always a rate of change: how much one quantity changes compared to another.

#220 · Algebra

A gym charges a \$30 sign-up fee plus a monthly cost. Which part is the y-intercept?

Answer

The \$30 sign-up fee is the y-intercept. It is what you pay before any months have passed, meaning when $x = 0$. The monthly cost would be the slope.

#221 · Algebra

In coding, what is a variable?

Answer

A variable in coding is a labeled storage spot for a number, similar to x and y in math. The difference is that in code its value can be updated as the program runs.

#222 · Algebra

In coding, what is a loop?

Answer

A loop is a set of instructions that repeats automatically, usually a fixed number of times or until some condition is met. For example, a loop can calculate y for $x = 0, 1, 2, 3, 4, 5$ one after another.

#223 · Algebra

In coding, what is an if-statement?

Answer

An if-statement is a rule that only runs certain instructions when a condition is true, and skips them otherwise. For example: if the calculated y equals the point's y -value, print that the point is on the line.

#224 · Algebra

Describe a simple program that builds a table of values for $y = 2x + 4$.

Answer

Set a variable x to 0. Loop while x is less than or equal to 5: calculate $y = 2x + 4$, record the pair (x, y) , then increase x by 1. Running it gives $(0,4)$, $(1,6)$, $(2,8)$, $(3,10)$, $(4,12)$, $(5,14)$.

#225 · Algebra

How would a program check if the point (3, 10) lies on $y = 2x + 4$?

Answer

Take the point's x-value, 3, and calculate $y = 2(3) + 4 = 10$. Then use an if-statement: if the calculated y equals the point's y-value, the point satisfies the relation. Since 10 equals 10, the point is on the line.

#226 · Data

What is categorical data?

Answer

Categorical data describes qualities or categories, not amounts. Examples are favorite pizza topping, eye color, or type of pet. You cannot do math like averaging with it. You can only count how often each category appears.

#227 · Data

What is numerical data?

Answer

Numerical data is made of actual numbers that represent a quantity, like height, test scores, or number of pets. It is also called quantitative data. It splits into two kinds: discrete and continuous.

#228 · Data

What is discrete data?

Answer

Discrete data can only take specific, separate values, usually whole numbers. The number of siblings a person has is discrete because you cannot have 2.5 siblings. You count discrete data rather than measure it.

#229 · Data

What is continuous data?

Answer

Continuous data can take any value in a range, including decimals. Height is continuous because someone could be 162.3 cm tall. Continuous data usually comes from measuring, not counting.

#230 · Data

Classify: favorite subject, number of pets, height in cm.

Answer

Favorite subject is categorical because it names a category. Number of pets is numerical and discrete because it can only be whole numbers. Height is numerical and continuous because it can take any decimal value.

#231 · Data

Is shoe size discrete or continuous?

Answer

Shoe size is discrete. Even though sizes include halves like 8.5, they can only take specific, separate values with nothing allowed in between. Continuous data can take any value in a range.

#232 · Data

Is the time to run 100 m discrete or continuous?

Answer

It is continuous. Time can take any value in a range, like 14.37 seconds. Measured quantities such as time, height, and mass are continuous.

#233 · Data

What is a census?

Answer

A census collects data from every single member of a population, meaning the whole group you care about. It gives complete information but takes a lot of time and money. That is why samples are often used instead.

#234 · Data

What is a sample?

Answer

A sample collects data from only part of the population. You then use that smaller group to make guesses about the whole. Sampling is faster and cheaper than a census, but it risks bias if the sample is unfair.

#235 · Data

What is bias in sampling?

Answer

Bias means the sample does not fairly represent the whole population, so the results are skewed in one direction. For example, asking only the basketball team which sport to fund will almost always produce basketball. A fair sample should be random and large enough.

#236 · Data

Why is surveying only cafeteria students at 11:00 am biased?

Answer

It misses students who eat lunch later or bring lunch from home. The sample only represents one slice of the school, not the whole population. A fair sample would include students from different grades, times, and habits.

#237 · Data

What makes a sample fair?

Answer

A fair sample should be chosen randomly, so every member of the population has a chance of being picked. It should also be large enough to represent different grades, interests, and backgrounds. Small or hand-picked samples produce biased results.

#238 · Data

What is a frequency table?

Answer

A frequency table organizes raw data by listing each possible value next to how many times it occurred. Frequency just means how often something happens. It turns a messy list of data into a clear two-column summary.

#239 · Data

What is a tally chart and how are marks grouped?

Answer

A tally chart records one mark each time a value appears in the data. Marks are grouped in bundles of five, with the fifth mark drawn as a diagonal line across the first four. This makes counting fast because you count by fives.

#240 · Data

How do you check a frequency table is correct?

Answer

Add up the whole frequency column. The total must match the number of data points you started with. If 20 students were surveyed, the frequencies must add to 20.

#241 · Data

**Frequencies are 2, 7, 6, 4, 1 for 20 students.
Which value is most common?**

Answer

The value with frequency 7 is the most common because 7 is the highest frequency. Also check the total: $2 + 7 + 6 + 4 + 1 = 20$, which matches the 20 students. That confirms the counting is correct.

#242 · Data

A tally shows 3 bundles of five plus 2 extra marks. What is the frequency?

Answer

Each bundle of five counts as 5, so 3 bundles equal 15. Add the 2 extra single marks to get 17. Counting in fives first is faster and safer than counting every mark.

#243 · Data

Trap: what is the most common tally chart mistake?

Answer

Miscounting the bundles of five and single marks. Always add up the final frequency column and check that the total matches the number of data points you collected. If it does not match, recount.

#244 · Data

When do you use a bar graph?

Answer

Use a bar graph for categorical data or discrete numerical data. Each separate bar shows the frequency of one category or value. The bars do not touch because the categories are separate.

#245 · Data

When do you use a histogram?

Answer

Use a histogram for continuous numerical data grouped into ranges called intervals or bins. An example is test scores grouped as 60 to 69, 70 to 79, and so on. The bars touch because the data is continuous.

#246 · Data

Key visual difference: bar graph vs histogram?

Answer

In a bar graph the bars are separate and do not touch, because each bar is its own category. In a histogram the bars touch with no gaps, because the intervals of continuous data flow into each other. Spotting gaps or no gaps tells you which graph you are looking at.

#247 · Data

When do you use a circle graph?

Answer

Use a circle graph, also called a pie graph, to show how a whole is split into parts. Each slice shows what percent of the total a category makes up. It works best when you want to emphasize each category's share of the whole.

#248 · Data

20 of 40 students walk to school. What percent is that?

Answer

Divide 20 by 40 to get 0.5, then multiply by 100 to get 50%. In a circle graph, walking would take half the circle. Always check that all category percents add to 100%.

#249 · Data

How do you turn a percent into a circle graph angle?

Answer

Write the percent as a decimal and multiply by 360, because a full circle has 360 degrees. For example, 25% is 0.25 times 360, which equals 90 degrees. All the slice angles together must add to 360 degrees.

#250 · Data

What angle does a 15% slice get in a circle graph?

Answer

Multiply 0.15 by 360 degrees to get 54 degrees. The percent as a decimal times 360 always gives the slice angle. Check your work by making sure all slices sum to 360.

#251 · Data

10 of 40 students take the bus. Find the percent and the slice angle.

Answer

10 divided by 40 equals 0.25, which is 25%. Multiply 0.25 by 360 to get a 90 degree slice. That is a quarter of the circle, which matches a quarter of the students.

#252 · Data

Which graph fits test scores grouped 60 to 69, 70 to 79, 80 to 89?

Answer

A histogram. The scores are continuous numerical data grouped into intervals, so the bars should touch with no gaps. A bar graph would be wrong because it is for separate categories.

#253 · Data

Which graph best shows each category's share of a whole?

Answer

A circle graph, because each slice's size shows the category's percent of the total. Use it when the message is about parts of a whole, like how students travel to school. Use bar graphs when comparing counts across categories instead.

#254 · Data

What is a measure of central tendency?

Answer

It is a single number that describes the middle or typical value of a data set. The three main measures are mean, median, and mode. Each is calculated differently and is useful in different situations.

#255 · Data

How do you calculate the mean?

Answer

Add up every value in the data set, then divide by how many values there are. The mean is what most people call the average. Because it uses every value, one extreme value can pull it up or down.

#256 · Data

How do you find the median?

Answer

First sort the data from smallest to largest. If there is an odd number of values, the median is the single middle value. If there is an even number, the median is the mean of the two middle values.

#257 · Data

What is the mode of a data set?

Answer

The mode is the value that appears most often. A data set can have one mode, more than one mode if values tie, or no mode at all if every value appears equally often. It is the only measure that works for categorical data.

#258 · Data

Find the median of 3, 7, 8, 12, 15.

Answer

The data is already sorted and has 5 values, an odd number. The median is the middle value, which is the 3rd one. So the median is 8.

#259 · Data

Find the mean of 4, 6, 8, 10, 12.

Answer

Add the values: $4 + 6 + 8 + 10 + 12 = 40$. Divide by the count of 5 values. The mean is 40 divided by 5, which equals 8.

#260 · Data

Find the median of 4, 8, 10, 14.

Answer

There are 4 values, an even number, so the median is the mean of the two middle values. The middle values are 8 and 10, and their sum is 18. Divide 18 by 2 to get a median of 9.

#261 · Data

Find the mode of 2, 3, 3, 5, 3, 7.

Answer

Count how often each value appears. The value 3 appears three times, more than any other value. So the mode is 3.

#262 · Data

Find the mean of 10, 20, 30, 40.

Answer

Add the values: $10 + 20 + 30 + 40 = 100$. Divide by the count of 4 values. The mean is 100 divided by 4, which equals 25.

#263 · Data

Find the median of 9, 2, 5, 7, 1.

Answer

First sort the data: 1, 2, 5, 7, 9. There are 5 values, so the median is the 3rd one. The median is 5, and sorting first is the step most students forget.

#264 · Data

What is an outlier?

Answer

An outlier is a value that sits far away from the rest of the data, either much higher or much lower. Outliers pull the mean toward them because the mean uses every value. The median barely moves, so it handles outliers better.

#265 · Data

When is the median better than the mean?

Answer

When the data set has an outlier. The outlier drags the mean toward it, so the mean stops representing a typical value. The median stays near the middle of the data, so it gives a fairer picture.

#266 · Data

Allowances: \$10, \$12, \$13, \$14, \$15, \$16, \$100. Why is the mean misleading?

Answer

The sum is 180, so the mean is 180 divided by 7, about \$25.70. But six of the seven students get between \$10 and \$16, so \$25.70 is not typical. The \$100 outlier pulled the mean up, and the median of \$14 is fairer.

#267 · Data

Can a data set have no mode or two modes?

Answer

Yes to both. If every value appears the same number of times, there is no mode. If two values tie for appearing most often, the set has two modes.

#268 · Data

Find the median of 1, 2, 3, 100.

Answer

There are 4 values, so average the two middle ones, 2 and 3. The median is 5 divided by 2, which is 2.5. Notice the outlier 100 does not affect the median at all.

#269 · Data

Which measure works for categorical data like favorite color?

Answer

Only the mode. You cannot add or order categories, so mean and median are impossible. The mode simply names the most frequent category, like the most popular color.

#270 · Data

Trap: mean vs median, what is the key difference in method?

Answer

Mean means add all the values and divide by the count. Median means sort the values first, then pick the middle one, or average the two middle values if the count is even. Always reread the question to check which one it wants.

#271 · Data

Scores 62, 75, 75, 80, 85, 90, 95: find mean, median, and mode.

Answer

The sum is 562, and 562 divided by 7 is about 80.3, the mean. With 7 sorted values the median is the 4th, which is 80. The mode is 75 because it appears twice and every other value appears once.

#272 · Data

How do you calculate the range of a data set?

Answer

Subtract the smallest value from the largest value. Range equals maximum minus minimum. It measures how spread out the data is with a single number.

#273 · Data

Find the range of 12, 5, 9, 20.

Answer

The largest value is 20 and the smallest is 5. The range is 20 minus 5, which equals 15. Always identify the true maximum and minimum before subtracting.

#274 · Data

What does a small range tell you? A large range?

Answer

A small range means the values are bunched close together, which is low variability. A large range means the values are spread far apart, which is high variability. Range gives a quick first look at spread.

#275 · Data

Trap: why can range be misleading?

Answer

Range only uses the two extreme values, so one outlier can inflate it. Set A: 10, 11, 12, 13, 14 has range 4, while Set B: 10, 11, 12, 13, 50 has range 40. Set B looks far more spread out, but only the one value 50 is unusual.

#276 · Data

What is two-variable data?

Answer

Two-variable data measures two different quantities on the same subjects, like each student's hours studied and test score. The goal is to see if there is a relationship between the two variables. One-variable data studies just a single quantity.

#277 · Data

What is a scatter plot?

Answer

A scatter plot displays two-variable data as points. Each point represents one subject, placed using one value on the x-axis and one on the y-axis. For example, one student could be plotted at (hours studied, test score).

#278 · Data

What is positive correlation?

Answer

Positive correlation means that as one variable increases, the other tends to increase too. The points on the scatter plot trend upward from left to right. An example is hours studied and test scores.

#279 · Data

What is negative correlation?

Answer

Negative correlation means that as one variable increases, the other tends to decrease. The points trend downward from left to right. An example is hours of gaming the night before a test and the test score.

#280 · Data

What does no correlation look like on a scatter plot?

Answer

The points are scattered randomly with no clear upward or downward pattern. Knowing one variable tells you nothing about the other. Do not confuse this with a weak correlation, where a faint trend still exists.

#281 · Data

Points: (1, 60), (2, 68), (3, 74), (4, 80), (5, 88), (6, 93). Correlation type?

Answer

As the x-values rise from 1 to 6, the y-values rise steadily from 60 to 93. This is a positive correlation. The points would trend upward from left to right on a scatter plot.

#282 · Data

Points: (1, 90), (2, 85), (3, 78), (4, 70), (5, 65), (6, 58). Correlation type?

Answer

As the x-values increase, the y-values steadily decrease from 90 to 58. This is a negative correlation. The points would trend downward from left to right.

#283 · Data

What is a line of best fit?

Answer

It is a single straight line drawn through a scatter plot to represent the overall trend. It usually does not pass through every point exactly. A good line has roughly the same number of points above it as below it.

#284 · Data

How do you use a line of best fit to make a prediction?

Answer

Find the x-value you are interested in, like 3.5 hours of studying. Read straight up to the line, then across to the y-axis to read the predicted value. This lets you estimate values that were never directly measured.

#285 • Data

Trap: does correlation prove causation?

Answer

No. Two variables can rise and fall together for another reason entirely. Ice cream sales and sunburns both rise in summer, but ice cream does not cause sunburns; hot sunny weather causes both. Remember: correlation does not equal causation.

#286 • Data

Trap: how do you judge a scatter plot's trend correctly?

Answer

Look at the overall direction of the whole cloud of points, not just two or three individual points. A few odd points do not erase a clear upward or downward trend. Confusing no correlation with a weak correlation is a common exam mistake.

#287 · Data

What is theoretical probability?

Answer

Theoretical probability is calculated with logic before any event happens. The formula is favorable outcomes divided by total possible outcomes. A favorable outcome is any outcome that counts as a success.

#288 · Data

What is experimental probability?

Answer

Experimental probability comes from actually performing trials and recording real results. The formula is times the event occurred divided by total trials. It usually lands close to, but not exactly on, the theoretical value.

#289 · Data

What is the probability of rolling a 4 on a standard die?

Answer

A die has 6 sides and only one shows a 4, so the probability is $\frac{1}{6}$. As a decimal that is about 0.167, and as a percent about 16.7%. This is a theoretical probability because it uses logic, not trials.

#290 · Data

A bag has 5 red, 3 blue, 2 green marbles. P(blue)?

Answer

There are 3 favorable outcomes out of 10 marbles in total. The probability is $\frac{3}{10}$, which is 0.3 as a decimal. As a percent, that is 30%.

#291 · Data

A die is rolled 50 times and shows a 4 exactly 8 times. Experimental probability?

Answer

Divide the 8 times the event occurred by the 50 total trials. That gives $8/50$, which equals 0.16, or 16%. It is close to the theoretical 16.7% but not exact, which is normal.

#292 · Data

What is the probability of rolling an even number on a die?

Answer

The even outcomes are 2, 4, and 6, which is 3 favorable outcomes out of 6. The probability is $3/6$, which simplifies to $1/2$. As a decimal that is 0.5, or 50%.

#293 · Data

Write the probability $\frac{3}{5}$ as a decimal and a percent.

Answer

Divide 3 by 5 to get 0.6. Multiply by 100 to get 60%. Every probability can be written as a fraction, a decimal, or a percent, and you should be able to switch between all three.

#294 · Data

What do probabilities of 0 and 1 mean?

Answer

A probability of 0 means the event is impossible. A probability of 1 means the event is certain. Every probability sits between 0 and 1, so an answer like 1.4 or a negative number must be a mistake.

#295 · Data

Why do experimental and theoretical probability usually differ?

Answer

Real trials involve chance, so results rarely match the theory exactly, especially with few trials. That difference is normal, not an error. The more trials you perform, the closer experimental probability tends to get to the theoretical value.

#296 · Data

Trap: a question describes 40 recorded coin flips. Which probability does it want?

Answer

Experimental probability, because it describes real trials that were actually recorded. Divide the number of times the event happened by the 40 trials. Theoretical probability is only for reasoning about outcomes before or without an experiment.

#297 · Data

What is a truncated axis and why is it misleading?

Answer

A truncated axis is a vertical axis that does not start at zero, for example starting at 50. Cutting out the bottom space makes small real differences between bars look huge. Always check where the axis starts before trusting how big a difference looks.

#298 · Data

What is cherry-picking data?

Answer

Cherry-picking means showing only the data or time period that supports your point while leaving out the rest. A company might advertise that sales are up this month while hiding a year-long decline. Always ask whether the graph shows the complete picture.

#299 · Data

Name two more graph tricks besides a truncated axis.

Answer

Inconsistent scales, where axis numbers jump unevenly, like 0 to 10 to 50 with equal spacing, distorting how steep a trend looks. Also 3D or oversized images, which make the eye judge visual size instead of the actual number. Both can make differences look bigger or smaller than they really are.

#300 · Data

What three things should you check on any graph?

Answer

First, does the vertical axis start at zero. Second, is the scale consistent, with equal jumps between numbers. Third, does the data show the complete picture, or just a cherry-picked slice that supports one side.

#301 · Geometry and Measurement

How many millimetres are in 1 centimetre?

Answer

There are 10 mm in 1 cm. To change centimetres into millimetres, multiply by 10. To change millimetres into centimetres, divide by 10.

#302 · Geometry and Measurement

How many centimetres are in 1 metre?

Answer

There are 100 cm in 1 m. Multiply by 100 to go from metres to centimetres. Divide by 100 to go the other way.

#303 · Geometry and Measurement

How many metres are in 1 kilometre?

Answer

There are 1000 m in 1 km. Multiply by 1000 to change kilometres into metres. Divide by 1000 to change metres into kilometres.

#304 · Geometry and Measurement

When converting units, do you multiply or divide going from a bigger unit to a smaller one?

Answer

You multiply when going from a bigger unit to a smaller unit, like metres to centimetres. You divide when going from a smaller unit to a bigger unit. Smaller units mean you need more of them, so the number gets bigger.

#305 · Geometry and Measurement

Convert 4.5 m to centimetres.

Answer

Since 1 m equals 100 cm, multiply by 100. So 4.5 times 100 equals 450 cm. Going from a bigger unit to a smaller unit always means multiplying.

#306 · Geometry and Measurement

Convert 3200 mm to metres.

Answer

There are 1000 mm in 1 m, so divide by 1000. That gives 3200 divided by 1000, which is 3.2 m. Going from a smaller unit to a bigger unit means dividing.

#307 · Geometry and Measurement

Convert 2.4 km to metres.

Answer

Since 1 km equals 1000 m, multiply by 1000. So 2.4 times 1000 equals 2400 m. Kilometres are bigger than metres, so the number of metres must be larger.

#308 · Geometry and Measurement

How many square centimetres are in 1 square metre?

Answer

There are 10 000 cm squared in 1 m squared, not 100. Area is length times length, so the conversion factor gets squared: 100 times 100 equals 10 000. Forgetting to square the factor is one of the most common exam mistakes.

#309 · Geometry and Measurement

How many square millimetres are in 1 square centimetre?

Answer

There are 100 mm squared in 1 cm squared. Since 1 cm equals 10 mm, a square centimetre is 10 mm by 10 mm, which is 100 mm squared. The length factor of 10 gets squared for area.

#310 · Geometry and Measurement

Convert 3.5 m squared to cm squared.

Answer

Multiply by 10 000, not 100, because area conversions use the length factor squared. So 3.5 times 10 000 equals 35 000 cm squared. Always ask whether you are converting a length or an area before choosing the factor.

#311 · Geometry and Measurement

How many cubic centimetres are in 1 cubic metre?

Answer

There are 1 000 000 cm cubed in 1 m cubed. Volume is length times length times length, so the factor of 100 gets cubed: 100 times 100 times 100 equals 1 000 000.

#312 · Geometry and Measurement

How many cubic millimetres are in 1 cubic centimetre?

Answer

There are 1000 mm cubed in 1 cm cubed. Since 1 cm equals 10 mm, a cubic centimetre is 10 by 10 by 10 mm, which is 1000 mm cubed. The length factor gets cubed for volume.

#313 · Geometry and Measurement

How many cubic centimetres equal 1 litre?

Answer

One litre equals 1000 cm cubed. This connects volume to capacity, which is how much liquid a container holds. To change cm cubed into litres, divide by 1000.

#314 · Geometry and Measurement

A fish tank has a volume of 45 000 cm cubed. How many litres does it hold?

Answer

Divide by 1000 because 1 L equals 1000 cm cubed. So 45 000 divided by 1000 equals 45 L. This conversion shows up often in tank, pool, and container problems.

#315 · Geometry and Measurement

Trap: why can you not use 100 to convert m squared to cm squared?

Answer

The factor 100 only works for plain lengths. Area is two lengths multiplied together, so the factor must be squared, giving 10 000. Using the plain length factor for area or volume is the most common conversion error on tests.

#316 · Geometry and Measurement

What is perimeter?

Answer

Perimeter is the total distance around the outside edge of a flat shape. You find it by adding up the length of every side. It is measured in plain length units like cm or m.

#317 · Geometry and Measurement

What is the formula for the perimeter of a rectangle?

Answer

P equals 2 times (length plus width), because a rectangle has two pairs of equal sides. For example, a rectangle 8 cm by 5 cm has P equals 2 times 13, which is 26 cm.

#318 · Geometry and Measurement

A triangle has sides 5 cm, 6 cm, and 7 cm. What is its perimeter?

Answer

There is no shortcut formula for a triangle, so just add all three sides. P equals 5 plus 6 plus 7, which is 18 cm.

#319 · Geometry and Measurement

How do you find the perimeter of an L-shaped composite figure?

Answer

Trace around the entire outer boundary and add every segment, including sides that are not labeled. Find missing sides by subtracting known lengths from the full width or height. Then add all the sides together.

#320 · Geometry and Measurement

What is circumference?

Answer

Circumference is the special name for the perimeter of a circle, the distance all the way around it. It is found using pi, which is about 3.14, times the diameter.

#321 · Geometry and Measurement

What is the circumference formula using the diameter?

Answer

C equals pi times d, where d is the diameter, the distance straight across the circle through its centre. Pi is a fixed number, about 3.14, describing how circumference relates to diameter.

#322 · Geometry and Measurement

What is the circumference formula using the radius?

Answer

C equals 2 times pi times r. This works because the diameter is double the radius, so pi times d becomes pi times 2r. Both forms give the same answer.

#323 · Geometry and Measurement

How are radius and diameter related?

Answer

The diameter is always exactly double the radius: d equals $2r$. The radius goes from the centre to the edge, while the diameter goes all the way across through the centre. Mixing these up is a classic exam trap.

#324 · Geometry and Measurement

A circle has a diameter of 14 cm. What is its circumference?

Answer

Use C equals π times d . That gives 3.14 times 14, which equals 43.96 cm, or about 44 cm. Since the diameter was given, you did not need to double anything.

#325 · Geometry and Measurement

A circle has a radius of 6 cm. What is its circumference?

Answer

Use C equals 2 times pi times r . That gives 2 times 3.14 times 6, which equals 37.68 cm. You could also double the radius to get diameter 12 and use pi times d .

#326 · Geometry and Measurement

What is the area formula for a rectangle?

Answer

A equals length times width. For example, a rectangle 12 cm long and 7 cm wide has area 12 times 7, which is 84 cm squared. Area is always given in square units.

#327 · Geometry and Measurement

What is the area formula for a triangle?

Answer

A equals one half times base times height. The height must be measured at a right angle from the base to the opposite corner, not along a slanted side.

#328 · Geometry and Measurement

**A triangle has base 10 cm and height 6 cm.
What is its area?**

Answer

A equals one half times 10 times 6. Half of 60 is 30, so the area is 30 cm squared. Remember to take half, or your answer will be double the true area.

#329 · Geometry and Measurement

Trap: which height do you use in triangle and parallelogram area formulas?

Answer

Always use the perpendicular height, the one that meets the base at a right angle. Never use a slanted side length as the height. Using the slant instead of the true height gives an answer that is too big.

#330 · Geometry and Measurement

What is the area formula for a parallelogram?

Answer

A equals base times height, using the perpendicular height, not the slanted side. Unlike a triangle, there is no one half in this formula.

#331 · Geometry and Measurement

A parallelogram has base 9 cm and height 4 cm. What is its area?

Answer

A equals base times height, so 9 times 4 equals 36 cm squared. The height here is the straight up and down distance between the two parallel bases.

#332 · Geometry and Measurement

What is the area formula for a trapezoid?

Answer

A equals (base1 plus base2) divided by 2, times height. A trapezoid has one pair of parallel sides, and those two sides are base1 and base2. You are averaging the two bases, then multiplying by the height.

#333 · Geometry and Measurement

A trapezoid has parallel sides 8 cm and 14 cm and height 5 cm. What is its area?

Answer

Average the bases first: 8 plus 14 equals 22, and 22 divided by 2 equals 11. Then multiply by the height: 11 times 5 equals 55 cm squared.

#334 · Geometry and Measurement

What is the area formula for a circle?

Answer

A equals pi times r squared. Square the radius first, then multiply by pi. This formula always needs the radius, never the diameter.

#335 · Geometry and Measurement

A circle has a radius of 7 cm. What is its area?

Answer

A equals pi times r squared, so 3.14 times 7 squared. Since 7 squared is 49, the area is 3.14 times 49, which equals 153.86 cm squared.

#336 · Geometry and Measurement

Trap: a circle's diameter is 10 cm. What is its area?

Answer

Do not square the diameter. Halve it first to get the radius: 10 divided by 2 equals 5 cm. Then A equals 3.14 times 5 squared, which is 3.14 times 25, giving 78.5 cm squared.

#337 · Geometry and Measurement

Which circle formula can use the diameter directly, and which cannot?

Answer

Circumference can use the diameter directly, since C equals π times d . Area cannot: A equals π times r squared always needs the radius. If a question gives the diameter for an area problem, cut it in half first.

#338 · Geometry and Measurement

How do you find the area of a semicircle?

Answer

Find the full circle's area with π times r squared, then take half of it. A semicircle is exactly half a circle, so its area is one half times π times r squared.

#339 · Geometry and Measurement

What does it mean to decompose a composite shape?

Answer

A composite shape combines two or more basic shapes, like an L-shaped room. Decomposing means mentally cutting it into simple pieces such as rectangles, triangles, trapezoids, or circles. You find each piece's area separately, then add them together.

#340 · Geometry and Measurement

A 10 by 6 cm rectangle has a triangle on top, base 10 cm, height 4 cm. Total area?

Answer

The rectangle's area is 10 times 6, which is 60 cm squared. The triangle's area is one half times 10 times 4, which is 20 cm squared. Adding them gives 60 plus 20, which equals 80 cm squared.

#341 · Geometry and Measurement

When do you subtract areas instead of adding them in a composite shape problem?

Answer

Subtract when a smaller shape has been removed from a larger one, like a square plate with a circular hole cut out. Find the big shape's area, find the removed shape's area, then subtract the hole from the whole.

#342 · Geometry and Measurement

A 12 by 12 cm square has a circular hole of radius 3 cm. Remaining area?

Answer

The square's area is 12 times 12, which is 144 cm squared. The circle's area is 3.14 times 3 squared, which is 3.14 times 9, giving 28.26 cm squared. Subtracting gives 144 minus 28.26, which equals 115.74 cm squared.

#343 · Geometry and Measurement

What is surface area, and what is a net?

Answer

Surface area is the total area of every outside face of a 3D solid added together. A net is the flat 2D pattern you get by unfolding the solid, showing all its faces laid out. Drawing the net helps you make sure no face gets missed.

#344 · Geometry and Measurement

What is the surface area formula for a rectangular prism?

Answer

SA equals 2 times (lw plus lh plus wh). A box has 6 rectangular faces in 3 matching pairs, so you find the three different face areas and double the total.

#345 · Geometry and Measurement

A rectangular prism is 5 cm by 4 cm by 3 cm. What is its surface area?

Answer

The three face areas are 5 times 4 equals 20, 5 times 3 equals 15, and 4 times 3 equals 12. Their sum is 47, and doubling gives SA equals 2 times 47, which is 94 cm squared.

#346 · Geometry and Measurement

What is the surface area formula for a triangular prism?

Answer

SA equals 2 times the triangle's area, plus the triangle's perimeter times the prism's length. The two triangles are the ends, and the perimeter times length covers the three rectangular sides that wrap around.

#347 · Geometry and Measurement

Triangular prism: end triangle area 12 cm squared, perimeter 16 cm, length 10 cm. Surface area?

Answer

The two triangle ends give 2 times 12, which is 24 cm squared. The wrap-around rectangles give 16 times 10, which is 160 cm squared. Adding gives 24 plus 160, which equals 184 cm squared.

#348 · Geometry and Measurement

What is the surface area formula for a cylinder?

Answer

SA equals 2 times pi times r squared, plus 2 times pi times r times h. The first part is the two circular ends, and the second part is the curved side, which unrolls into a rectangle whose width equals the circumference.

#349 · Geometry and Measurement

A cylinder has radius 3 cm and height 10 cm. What is its surface area?

Answer

The two circles give 2 times 3.14 times 9, which is 56.52 cm squared. The curved side gives 2 times 3.14 times 3 times 10, which is 188.4 cm squared. Adding gives 244.92 cm squared.

#350 · Geometry and Measurement

What is the surface area formula for a cone?

Answer

SA equals pi times r squared, plus pi times r times l, where l is the slant height. The first part is the circular base and the second part is the curved side. The slant height is measured along the outside surface from the base edge to the tip.

#351 · Geometry and Measurement

A cone has radius 4 cm and slant height 9 cm. What is its surface area?

Answer

The base gives 3.14 times 16, which is 50.24 cm squared. The curved side gives 3.14 times 4 times 9, which is 113.04 cm squared. Adding gives 163.28 cm squared.

#352 · Geometry and Measurement

Trap: what is the difference between slant height and true height in a cone or pyramid?

Answer

The true height is the straight up and down distance from the base to the tip. The slant height runs along the outside surface from the base edge to the tip, so it is always longer. Surface area formulas use the slant height, but volume formulas use the true height.

#353 · Geometry and Measurement

What is the surface area formula for a square based pyramid?

Answer

SA equals the base area, plus 4 times (one half times base side times slant height). The net is one square base plus 4 identical triangular faces, so you add the square's area to the four triangle areas.

#354 · Geometry and Measurement

A square pyramid has base side 6 cm and slant height 5 cm. What is its surface area?

Answer

The base area is 6 times 6, which is 36 cm squared. Each triangular face is one half times 6 times 5, which is 15 cm squared, and four of them give 60 cm squared. Adding gives 36 plus 60, which equals 96 cm squared.

#355 · Geometry and Measurement

What is the surface area formula for a sphere?

Answer

SA equals 4 times pi times r squared. A sphere is a perfectly round ball with no flat faces and no true net, but this formula still gives its total outside area.

#356 · Geometry and Measurement

A sphere has radius 5 cm. What is its surface area?

Answer

SA equals 4 times 3.14 times 5 squared. Since 5 squared is 25, that gives 4 times 3.14 times 25, which equals 314 cm squared.

#357 · Geometry and Measurement

Trap: what is a common mistake when finding the surface area of a solid?

Answer

Forgetting a face, like leaving out one of the two circular ends of a cylinder or a face of a box. Before calculating, list every face the solid has, drawing the net if it helps. Check your total face count against your list so nothing gets skipped.

#358 · Geometry and Measurement

What is volume, and what units is it measured in?

Answer

Volume measures how much space a 3D solid takes up, or how much it could hold inside. It is always given in cubic units, like cm cubed or m cubed, because you are filling a three dimensional space.

#359 · Geometry and Measurement

What is the volume formula for a rectangular prism?

Answer

V equals length times width times height. For example, a box 5 cm by 4 cm by 3 cm has volume 5 times 4 times 3, which equals 60 cm cubed.

#360 · Geometry and Measurement

What is the volume formula for a triangular prism?

Answer

V equals the area of the triangular end times the prism's length. For example, if the end triangle has area 12 cm squared and the prism is 10 cm long, the volume is 12 times 10, which equals 120 cm cubed.

#361 · Geometry and Measurement

What is the volume formula for a cylinder?

Answer

V equals pi times r squared times h. You are finding the circular base's area first, then multiplying by the height, just like any prism-style solid.

#362 · Geometry and Measurement

A cylinder has radius 3 cm and height 10 cm. What is its volume?

Answer

V equals 3.14 times 3 squared times 10. Since 3 squared is 9, that gives 3.14 times 90, which equals 282.6 cm cubed.

#363 · Geometry and Measurement

What is the volume formula for a cone?

Answer

V equals one third times pi times r squared times h, using the true vertical height, not the slant height. A cone holds exactly one third the volume of a cylinder with the same radius and height.

#364 · Geometry and Measurement

A cone has radius 4 cm and true height 9 cm. What is its volume?

Answer

V equals one third times 3.14 times 16 times 9. First, 3.14 times 16 times 9 equals 452.16, and one third of that is 150.72 cm cubed. Notice the volume formula uses the true height, not the slant height.

#365 · Geometry and Measurement

How does a cone's volume compare to a cylinder's with the same radius and height?

Answer

A cone always holds exactly one third the volume of that cylinder. So if a cylinder holds 300 cm cubed, a cone with the same base and height holds 100 cm cubed. The one third in the cone formula comes from this relationship.

#366 · Geometry and Measurement

What is the volume formula for a square based pyramid?

Answer

V equals one third times base area times height, using the straight up and down height. A pyramid holds exactly one third the volume of a rectangular prism with the same base and height.

#367 · Geometry and Measurement

A square pyramid has base side 6 cm and height 8 cm. What is its volume?

Answer

The base area is 6 times 6, which is 36 cm squared. Then V equals one third times 36 times 8, and one third of 288 is 96 cm cubed.

#368 · Geometry and Measurement

What is the volume formula for a sphere?

Answer

V equals four thirds times pi times r cubed. Cube the radius first, then multiply by pi, then by four thirds. Do not confuse this with the surface area formula, which uses r squared.

#369 · Geometry and Measurement

A sphere has radius 5 cm. What is its volume?

Answer

V equals four thirds times 3.14 times 5 cubed. Since 5 cubed is 125, that gives four thirds times 392.5, which equals about 523.33 cm cubed.

#370 · Geometry and Measurement

Trap: in a triangular prism, what two different things can the word height mean?

Answer

It can mean the height of the triangular end, used inside the triangle area formula. It can also mean the length of the whole prism, used for the side rectangles and the volume. Always label which height you are using before calculating.

#371 · Geometry and Measurement

For a fixed perimeter, which rectangle has the maximum area?

Answer

The square. With the perimeter locked at one value, the closer a rectangle is to a square, the bigger its area. Stretched, skinny rectangles waste perimeter and cover less area.

#372 · Geometry and Measurement

You have exactly 24 m of fencing for a rectangular yard. What shape gives the biggest area?

Answer

A 6 m by 6 m square, which uses all 24 m of fencing and gives 36 m squared. An 8 by 4 rectangle uses the same fencing but only gives 32 m squared. The square always wins for a fixed perimeter.

#373 · Geometry and Measurement

Why does a 10 by 2 m rectangle have less area than a 6 by 6 square?

Answer

Its perimeter is still 24 m, but its area is only 10 times 2, which is 20 m squared, compared to 36 m squared for the 6 by 6 square. As a rectangle gets more stretched out, the area shrinks even though the perimeter stays the same.

#374 · Geometry and Measurement

For a fixed area, which rectangle has the minimum perimeter?

Answer

The square again. If the area is locked at one value, the square needs the least perimeter to enclose it. Any longer, skinnier rectangle with the same area needs more perimeter.

#375 · Geometry and Measurement

Area fixed at 36 m squared: compare perimeters of a 6 by 6 square and 9 by 4 rectangle.

Answer

Both have area 36 m squared. The square's perimeter is 2 times (6 plus 6), which is 24 m. The 9 by 4 rectangle needs 2 times (9 plus 4), which is 26 m, so the square uses less fencing for the same area.

#376 · Geometry and Measurement

What does optimization mean in measurement problems?

Answer

Optimization means finding the best possible option out of many and using math to prove it. In Grade 9, it usually means showing a square gives the most area for a fixed perimeter, or the least perimeter for a fixed area.

#377 · Geometry and Measurement

What is the Cartesian plane?

Answer

It is a flat grid used to describe the exact location of a point using two numbers. It is named after the mathematician Rene Descartes. It is the same grid used for graphing in math class.

#378 · Geometry and Measurement

Which axis is the x-axis and which is the y-axis?

Answer

The x-axis is the horizontal number line that goes left and right. The y-axis is the vertical number line that goes up and down. They cross at a right angle.

#379 · Geometry and Measurement

What is the origin?

Answer

The origin is the point where the x-axis and y-axis cross. Its coordinates are $(0, 0)$. Every point on the plane is located by counting from the origin.

#380 · Geometry and Measurement

In the ordered pair (x, y) , what does each number tell you?

Answer

The x-value tells you how far to move left or right from the origin, with positive meaning right and negative meaning left. The y-value tells you how far to move up or down, with positive meaning up and negative meaning down. The x-value always comes first.

#381 · Geometry and Measurement

How do you plot the point $(-2, 5)$?

Answer

Start at the origin. Move 2 units left because the x-value is negative. Then move 5 units up because the y-value is positive.

#382 · Geometry and Measurement

What are the signs of x and y in Quadrant I?

Answer

In Quadrant I, the top right region, both x and y are positive. A point like $(4, 3)$ sits in Quadrant I. The quadrants are numbered counterclockwise starting from the top right.

#383 · Geometry and Measurement

What are the sign patterns in Quadrants II, III, and IV?

Answer

Quadrant II is top left: x is negative and y is positive. Quadrant III is bottom left: both x and y are negative. Quadrant IV is bottom right: x is positive and y is negative.

#384 · Geometry and Measurement

A point is in Quadrant III. What do you know about its coordinates before plotting?

Answer

Both coordinates must be negative, because Quadrant III is the bottom left region. Knowing the quadrant sign patterns lets you check your plotting work quickly. If your plotted point has a positive coordinate, something went wrong.

#385 · Geometry and Measurement

Where does the distance formula come from?

Answer

It comes from the Pythagorean theorem. The horizontal difference in x-values and the vertical difference in y-values form the two legs of a right triangle. The direct line between the points is the hypotenuse.

#386 · Geometry and Measurement

What is the formula for the distance between two points?

Answer

Distance equals the square root of (difference in x) squared plus (difference in y) squared. Find each difference, square both, add them, then take the square root of the sum.

#387 · Geometry and Measurement

Find the distance between A (2, 3) and B (7, 15).

Answer

The horizontal difference is 7 minus 2, which is 5, and the vertical difference is 15 minus 3, which is 12. Squaring gives 25 and 144, and their sum is 169. The square root of 169 is 13, so the distance is 13 units.

#388 · Geometry and Measurement

Find the distance between C (-1, 2) and D (4, -10).

Answer

The horizontal difference is 4 minus (-1), which is 5, and the vertical difference is -10 minus 2, which is -12. Squaring gives 25 and 144, since squaring a negative always gives a positive. The sum is 169, and its square root is 13 units.

#389 · Geometry and Measurement

Trap: what step do students often skip at the end of a distance formula problem?

Answer

Taking the square root. Squaring and adding the differences only gives the distance squared, not the distance itself. Always finish with the square root, and remember distance is always a positive value.

#390 · Geometry and Measurement

What is the midpoint of a line segment, and how do you find it?

Answer

The midpoint is the exact point halfway between the two endpoints. Add the two x-coordinates and divide by 2, then add the two y-coordinates and divide by 2. You are simply averaging the x-values and averaging the y-values.

#391 · Geometry and Measurement

Find the midpoint of E (2, 3) and F (8, 9).

Answer

Average the x-values: 2 plus 8 is 10, and half of 10 is 5. Average the y-values: 3 plus 9 is 12, and half of 12 is 6. The midpoint is (5, 6).

#392 · Geometry and Measurement

Find the midpoint of G (-4, 6) and H (2, -2).

Answer

Average the x-values: -4 plus 2 is -2, and -2 divided by 2 is -1. Average the y-values: 6 plus (-2) is 4, and 4 divided by 2 is 2. The midpoint is (-1, 2).

#393 · Geometry and Measurement

State the Pythagorean theorem.

Answer

In every right triangle, $a^2 + b^2 = c^2$. Here a and b are the two shorter sides, called legs, and c is the hypotenuse. It only works in triangles that have a 90 degree angle.

#394 · Geometry and Measurement

What is the hypotenuse and how do you find it in a diagram?

Answer

The hypotenuse is the longest side of a right triangle. It is always directly across from the right angle and never touches the right angle corner. Find the right angle first, then look for the side opposite it.

#395 · Geometry and Measurement

**A right triangle has legs 6 cm and 8 cm.
What is the hypotenuse?**

Answer

Square each leg: 6 squared is 36 and 8 squared is 64. Add them: 36 plus 64 is 100. The square root of 100 is 10, so the hypotenuse is 10 cm.

#396 · Geometry and Measurement

**A right triangle has hypotenuse 13 m and
one leg 5 m. What is the missing leg?**

Answer

Square the hypotenuse: 13 squared is 169. Square the known leg: 5 squared is 25. Subtract: 169 minus 25 is 144, and the square root of 144 is 12, so the missing leg is 12 m. To find a leg, you subtract instead of adding.

#397 · Geometry and Measurement

**A 10 foot ladder's base is 6 feet from a wall.
How high does it reach?**

Answer

The ladder is the hypotenuse, so square it: 10 squared is 100. Square the known leg: 6 squared is 36. Subtract: 100 minus 36 is 64, and the square root of 64 is 8, so the ladder reaches 8 feet up the wall.

#398 · Geometry and Measurement

**A park is 30 m wide and 40 m long. How
long is the diagonal path across it?**

Answer

The diagonal is the hypotenuse of a right triangle with legs 30 and 40. Squaring gives 900 and 1600, which add to 2500, and the square root of 2500 is 50 m. That beats walking 70 m around the two sides.

#399 · Geometry and Measurement

A TV screen is 16 inches wide and 12 inches tall. What is its diagonal size?

Answer

TV sizes are measured corner to corner, so the diagonal is the hypotenuse. Squaring gives 256 and 144, which add to 400, and the square root of 400 is 20. It is a 20 inch TV.

#400 · Geometry and Measurement

Trap: how do students pick the wrong hypotenuse?

Answer

They grab any slanted side without checking the right angle first. A slanted side that touches the right angle corner is a leg, not the hypotenuse. The hypotenuse is always the side directly across from the right angle.

#401 · Geometry and Measurement

Trap: what does stopping at a squared plus b squared give you?

Answer

It only gives you c^2 , not c . Many students hand in the squared value, like 100 instead of 10, because they forgot the final square root. Always take the square root as the last step.

#402 · Geometry and Measurement

Does the Pythagorean theorem work in every triangle?

Answer

No, it only works in right triangles, ones with exactly one 90 degree angle. If a triangle has no right angle, $a^2 + b^2$ will not equal c^2 . Check for the right angle before using the theorem.

#403 · Geometry and Measurement

What are complementary angles?

Answer

Two angles that add up to 90 degrees. If one angle is 35 degrees, its complement is 90 minus 35, which is 55 degrees. Picture them fitting together into one right angle.

#404 · Geometry and Measurement

What are supplementary angles?

Answer

Two angles that add up to 180 degrees. If one angle is 110 degrees, its supplement is 180 minus 110, which is 70 degrees. Picture them fitting together along one straight line.

#405 · Geometry and Measurement

What are opposite (vertical) angles?

Answer

When two straight lines cross, the two angles directly across from each other are called opposite or vertical angles. They are always equal. The angles next to each other along either line add to 180 degrees instead.

#406 · Geometry and Measurement

Two lines cross and one angle is 65 degrees. What are the other three angles?

Answer

The angle directly opposite is also 65 degrees, because opposite angles are equal. The two angles beside it each measure 180 minus 65, which is 115 degrees. All four angles together add up to 360 degrees.

#407 · Geometry and Measurement

What do angles on a straight line add up to?

Answer

They always add up to 180 degrees. If a line has angles of 40 degrees, 75 degrees, and one unknown, the unknown is $180 \text{ minus } 40 \text{ minus } 75$, which is 65 degrees.

#408 · Geometry and Measurement

What do angles around a single point add up to?

Answer

They always add up to 360 degrees. If four angles meet at a point and three measure 90, 80, and 100 degrees, the fourth is $360 \text{ minus } 90 \text{ minus } 80 \text{ minus } 100$, which is 90 degrees.

#409 · Geometry and Measurement

What is a transversal?

Answer

A transversal is a third line that crosses two parallel lines. Parallel lines never meet and stay the same distance apart. The transversal creates special angle pairs at the two intersections.

#410 · Geometry and Measurement

What are corresponding angles on parallel lines?

Answer

They are angles sitting in the same position at each intersection, like both in the top right corner. When the two lines are parallel, corresponding angles are always equal.

#411 · Geometry and Measurement

What are alternate interior angles on parallel lines?

Answer

They sit between the two parallel lines but on opposite sides of the transversal. When the lines are parallel, alternate interior angles are always equal. Many people spot them by their Z shape pattern.

#412 · Geometry and Measurement

What are co-interior angles on parallel lines?

Answer

They sit between the two parallel lines on the same side of the transversal. They are not equal; instead they always add up to 180 degrees. They are also called same-side interior angles.

#413 · Geometry and Measurement

A transversal crosses parallel lines and one angle is 72 degrees. Find the three related angles.

Answer

The corresponding angle is equal, so it is 72 degrees. The alternate interior angle is also equal, 72 degrees. The co-interior angle adds with it to 180, so it is 180 minus 72, which is 108 degrees.

#414 · Geometry and Measurement

Trap: how do you avoid mixing up alternate interior and co-interior angles?

Answer

Use this trick: if the two angles between the parallel lines are on the same side of the transversal, they add up to 180 degrees. If they are on opposite sides, they are equal. Same side means sum, opposite side means equal.

#415 · Geometry and Measurement

What do the interior angles of a triangle add up to?

Answer

Always exactly 180 degrees, no matter the triangle's shape. If a triangle has angles of 50 and 65 degrees, the third angle is 180 minus 50 minus 65, which is 65 degrees.

#416 · Geometry and Measurement

What are scalene, isosceles, and equilateral triangles?

Answer

A scalene triangle has three sides of different lengths. An isosceles triangle has exactly two equal sides, and the angles opposite those sides are also equal. An equilateral triangle has all three sides equal.

#417 · Geometry and Measurement

What are the angles in an equilateral triangle?

Answer

All three angles are 60 degrees each. Since every triangle's angles total 180 degrees and all three are equal, each one is 180 divided by 3, which is 60.

#418 · Geometry and Measurement

In an isosceles triangle, which angles are equal?

Answer

The two angles opposite the two equal sides are equal. So if you know one of those base angles, you automatically know the other. The third angle can then be found by subtracting both from 180.

#419 · Geometry and Measurement

What are acute, right, and obtuse triangles, and why can a triangle never have two right angles?

Answer

An acute triangle has all angles under 90 degrees, a right triangle has exactly one 90 degree angle, and an obtuse triangle has exactly one angle over 90 degrees. Two right angles alone would total 180 degrees, leaving nothing for the third angle. That is why two right or two obtuse angles are impossible.

#420 · Geometry and Measurement

What do the interior angles of a quadrilateral add up to, and why?

Answer

They always add up to 360 degrees. Drawing one diagonal splits any quadrilateral into two triangles, and each triangle holds 180 degrees. Two triangles together give 360 degrees.

#421 · Geometry and Measurement

A quadrilateral has angles of 100, 85, and 95 degrees. What is the fourth angle?

Answer

The four angles must total 360 degrees. So the fourth angle is 360 minus 100 minus 85 minus 95, which is 80 degrees.

#422 · Geometry and Measurement

What are the key properties of a parallelogram?

Answer

Both pairs of opposite sides are parallel and equal in length. Opposite angles are equal. Consecutive angles, the ones next to each other, are supplementary, meaning they add up to 180 degrees.

#423 · Geometry and Measurement

What makes a rectangle special among parallelograms?

Answer

A rectangle is a parallelogram where all four angles are exactly 90 degrees. It keeps every parallelogram property. It also adds a new rule: its two diagonals are equal in length.

#424 · Geometry and Measurement

What makes a rhombus special among parallelograms?

Answer

A rhombus is a parallelogram where all four sides are equal in length. It keeps every parallelogram property. Its diagonals cross at a right angle and cut each other exactly in half.

#425 · Geometry and Measurement

Why is a square both a rectangle and a rhombus?

Answer

A square has four 90 degree angles, which makes it a rectangle. It also has four equal sides, which makes it a rhombus. Since it meets both definitions at once, it inherits every property of both shapes.

#426 · Geometry and Measurement

Trap: can you use parallelogram rules on any quadrilateral?

Answer

No. Only use rules like equal opposite sides and equal opposite angles when the shape is actually stated or shown to be a parallelogram, rectangle, rhombus, or square. A general quadrilateral only guarantees its four angles add up to 360 degrees, nothing more.

#427 · Geometry and Measurement

What is a transformation, and what are the three basic types?

Answer

A transformation moves a shape to a new position on the grid without changing its size or shape. The three basic types are translations (slides), reflections (flips), and rotations (turns).

#428 · Geometry and Measurement

How do you translate a point on the Cartesian plane?

Answer

Add or subtract from its coordinates. Moving right adds to x and moving left subtracts from x . Moving up adds to y and moving down subtracts from y .

#429 · Geometry and Measurement

Translate the point (1, 4) by 3 units right and 2 units down.

Answer

Add 3 to the x-value: 1 plus 3 is 4. Subtract 2 from the y-value: 4 minus 2 is 2. The new point is (4, 2).

#430 · Geometry and Measurement

What is the coordinate rule for reflecting a point over the x-axis?

Answer

Keep the x-value the same and flip the sign of the y-value. For example, reflecting (5, 3) over the x-axis gives (5, -3). The point jumps to the mirror position below the horizontal axis.

#431 · Geometry and Measurement

What is the coordinate rule for reflecting a point over the y-axis?

Answer

Keep the y-value the same and flip the sign of the x-value. For example, reflecting (5, 3) over the y-axis gives (-5, 3). The point jumps to the mirror position on the other side of the vertical axis.

#432 · Geometry and Measurement

What is the coordinate rule for a 180 degree rotation around the origin?

Answer

Flip the sign of both coordinates. For example, rotating (4, -2) by 180 degrees gives (-4, 2). The point ends up directly across the origin from where it started.

#433 · Geometry and Measurement

What is the rule for rotating a point 90 degrees counterclockwise around the origin?

Answer

The rule turns (x, y) into $(-y, x)$. For example, rotating $(3, 5)$ by 90 degrees counterclockwise gives $(-5, 3)$. The new x is the old y with its sign flipped, and the new y is the old x .

#434 · Geometry and Measurement

What is the rule for rotating a point 90 degrees clockwise around the origin?

Answer

The rule turns (x, y) into $(y, -x)$. For example, rotating $(3, 5)$ by 90 degrees clockwise gives $(5, -3)$. It is the reverse of the counterclockwise rule, so applying one then the other brings the point back.

#435 · Geometry and Measurement

Trap: what is the most common transformation mistake?

Answer

Sign errors: forgetting to flip a sign during a reflection, or adding instead of subtracting during a translation. Either mistake lands the point in the wrong quadrant. After transforming, check that the new point sits in a quadrant that matches the direction you moved or flipped.

#436 · Financial Literacy

What is the simple interest formula?

Answer

The formula is $I = Prt$. I is the interest earned or owed, P is the principal (starting amount), r is the annual interest rate written as a decimal, and t is the time in years. Simple interest is always calculated on the original principal, which never changes.

#437 · Financial Literacy

In $I = Prt$, what does P stand for?

Answer

P is the principal, which is the original amount of money you deposit, invest, or borrow. In simple interest, the principal stays the same the whole time. Interest is always calculated on this starting amount, not on a growing balance.

#438 · Financial Literacy

In $I = Prt$, what does r stand for, and how do you write it?

Answer

The r is the annual interest rate written as a decimal. To convert a percent to a decimal, divide by 100. So 5% becomes 0.05 and 12% becomes 0.12. Forgetting this conversion is a very common test mistake.

#439 · Financial Literacy

In $I = Prt$, what does t stand for?

Answer

The t is time measured in years. If a question gives months, convert to years first. For example, 6 months is 0.5 years and 18 months is 1.5 years. Using months directly in the formula gives an answer that is 12 times too big.

#440 · Financial Literacy

You invest \$500 at 4% simple interest for 3 years. How much interest do you earn?

Answer

Use $I = Prt$ with $P = 500$, $r = 0.04$, and $t = 3$.
Multiply: $500 \times 0.04 = 20$, then $20 \times 3 = 60$. You earn \$60 in interest.

#441 · Financial Literacy

You deposit \$800 at 5% simple interest for 3 years. Find the total amount.

Answer

First find the interest: $I = 800 \times 0.05 \times 3 = \120 .
Then add it to the principal: $800 + 120 = \$920$.
Always check whether the question wants the interest alone or the total amount.

#442 · Financial Literacy

How do you find the total amount with simple interest?

Answer

Calculate the interest first using $I = Prt$, then add it to the principal. Total amount = $P + I$. Many students correctly find the interest but forget to add the principal back, which loses easy marks.

#443 · Financial Literacy

You borrow \$200 at 3% simple interest for 5 years. How much interest do you owe?

Answer

Use $I = Prt$ with $P = 200$, $r = 0.03$, and $t = 5$.
Multiply: $200 \times 0.03 = 6$, then $6 \times 5 = 30$. You owe \$30 in interest, so you would pay back \$230 in total.

#444 · Financial Literacy

\$100 earns 8% simple interest for 6 months. How much interest?

Answer

Convert 6 months to years first: $t = 0.5$. Then $I = 100 \times 0.08 \times 0.5 = \4 . If you had used $t = 6$ by mistake, you would get \$48, which is 12 times too big.

#445 · Financial Literacy

\$1000 earns \$120 in simple interest at 4% per year. How many years did it take?

Answer

Rearrange $I = Prt$ to solve for t . Here $t = I$ divided by (Pr) , so $t = 120 / (1000 \times 0.04) = 120 / 40 = 3$. It took 3 years.

#446 · Financial Literacy

\$600 earns \$90 in simple interest over 5 years. What is the rate?

Answer

Rearrange $I = Prt$ to solve for r . Here $r = I$ divided by (Pt) , so $r = 90 / (600 \times 5) = 90 / 3000 = 0.03$. The rate is 3% per year.

#447 · Financial Literacy

What principal earns \$50 at 5% simple interest in 2 years?

Answer

Rearrange $I = Prt$ to solve for P . Here $P = I$ divided by (rt) , so $P = 50 / (0.05 \times 2) = 50 / 0.1 = 500$. The principal is \$500.

#448 · Financial Literacy

What is the compound interest formula?

Answer

The formula is $A = P(1 + r/n)^{nt}$. A is the final amount, P is the principal, r is the annual rate as a decimal, n is how many times interest compounds per year, and t is the time in years. The bracket is raised to an exponent, which is what makes the money grow faster and faster.

#449 · Financial Literacy

In $A = P(1 + r/n)^{nt}$, what does A stand for?

Answer

A is the final amount, which means the principal plus all the interest that has been added. It is not just the interest. If a question asks for the interest earned, subtract the principal: interest = $A - P$.

#450 · Financial Literacy

In the compound interest formula, what does n stand for?

Answer

The n is the number of times interest is compounded per year. Annually means $n = 1$, semi-annually means $n = 2$, quarterly means $n = 4$, and monthly means $n = 12$. It appears twice in the formula: dividing the rate (r/n) and multiplying the time in the exponent (nt).

#451 · Financial Literacy

What is compound interest in plain words?

Answer

Compound interest means the interest itself starts earning interest. Each time interest is added, the next calculation uses the new, bigger balance instead of the original principal. It works like a snowball rolling downhill, picking up more snow the bigger it gets.

#452 · Financial Literacy

You invest \$800 at 5% compounded annually for 3 years. Find the final amount.

Answer

Use $A = 800(1.05)^3$. First $1.05 \times 1.05 = 1.1025$, then $1.1025 \times 1.05 = 1.157625$. Multiply by the principal: $800 \times 1.157625 = \$926.10$.

#453 · Financial Literacy

You invest \$1000 at 6% compounded annually for 10 years. Find the final amount.

Answer

Use $A = 1000(1.06)^{10}$. The value of 1.06^{10} is about 1.790847. So $A = 1000 \times 1.790847 = \1790.85 , rounded to the nearest cent.

#454 · Financial Literacy

You invest \$400 at 5% compounded annually for 2 years. Find the final amount.

Answer

Use $A = 400(1.05)^2$. First $1.05 \times 1.05 = 1.1025$. Then $400 \times 1.1025 = \$441$. The interest earned is $441 - 400 = \$41$.

#455 · Financial Literacy

You invest \$2000 at 4% compounded annually for 3 years. Find the final amount.

Answer

Use $A = 2000(1.04)^3$. Build up the power: $1.04 \times 1.04 = 1.0816$, then $1.0816 \times 1.04 = 1.124864$. Multiply: $2000 \times 1.124864 = \$2249.73$, rounded to the nearest cent.

#456 · Financial Literacy

You invest \$1000 at 10% compounded annually for 2 years. Find the final amount.

Answer

Use $A = 1000(1.10)^2$. First $1.10 \times 1.10 = 1.21$. Then $1000 \times 1.21 = \$1210$. Notice the second year earned \$110, more than the first year's \$100, because the interest earned interest.

#457 · Financial Literacy

Interest is 12% per year, compounded monthly. What rate and exponent go in the formula?

Answer

Divide the annual rate by 12: $r/n = 0.12/12 = 0.01$, which is 1% per month. Multiply the years by 12 for the exponent: $nt = 12t$. So for 1 year the formula uses $(1.01)^{12}$.

#458 · Financial Literacy

\$1000 at 12% compounded monthly for 1 year. Find the final amount.

Answer

Monthly rate is $0.12/12 = 0.01$, and the exponent is $12 \times 1 = 12$. So $A = 1000(1.01)^{12}$. Since 1.01^{12} is about 1.126825, the final amount is about \$1126.83. That beats \$1120 from annual compounding because of the extra compounding periods.

#459 · Financial Literacy

\$1000 at 6% compounded semi-annually for 1 year. Find the final amount.

Answer

Semi-annually means $n = 2$, so the rate per period is $0.06/2 = 0.03$ and the exponent is $2 \times 1 = 2$. $A = 1000(1.03)^2 = 1000 \times 1.0609 = \1060.90 . With annual compounding it would be only \$1060, so compounding more often earns slightly more.

#460 · Financial Literacy

How do you find just the interest earned from a compound interest problem?

Answer

The formula $A = P(1 + r/n)^{nt}$ gives the final amount, not the interest. To get the interest, subtract the principal: $\text{interest} = A - P$. For example, if \$800 grows to \$926.10, the interest earned is $926.10 - 800 = \$126.10$.

#461 · Financial Literacy

Trap: a question says interest is compounded monthly. Which formula do you use?

Answer

Any mention of compounding, whether annually, monthly, or quarterly, means you must use the compound formula $A = P(1 + r/n)^{nt}$. Using $I = Prt$ on a compounding question is one of the most common exam mistakes. Simple interest never mentions compounding.

#462 · Financial Literacy

Trap: what happens if you forget to divide the rate by n for monthly compounding?

Answer

You will use the full annual rate 12 times a year instead of one twelfth of it each month, so your answer comes out far too large. Always divide the annual rate by n to get the rate per period, and multiply the years by n to get the exponent. Both adjustments must happen together.

#463 · Financial Literacy

How is simple interest different from compound interest?

Answer

Simple interest is always calculated on the original principal, so it adds the same amount every year. Compound interest is calculated on the current balance, which includes past interest, so it grows faster and faster. Over short times the difference is small, but over many years it becomes huge.

#464 · Financial Literacy

\$800 at 5% for 3 years: how much more does compound earn than simple?

Answer

Simple interest gives $800 + (800 \times 0.05 \times 3) = \920 .
Compound interest gives $800(1.05)^3 = \$926.10$.
The difference is \$6.10, which is the extra interest earned on the interest itself.

#465 · Financial Literacy

\$1000 at 6% for 10 years: compare simple and compound totals.

Answer

Simple: $I = 1000 \times 0.06 \times 10 = \600 , so the total is \$1600. Compound: $A = 1000(1.06)^{10} = \$1790.85$. Compounding earns \$190.85 more, and the gap keeps widening the longer the money sits.

#466 · Financial Literacy

Why does the gap between simple and compound interest grow over time?

Answer

Simple interest adds the same fixed amount every year because it only uses the original principal. Compound interest adds a bigger amount each year because the balance it works from keeps growing. Each year of compounding builds on all the previous years, so the gap widens faster and faster.

#467 · Financial Literacy

Which type of interest do long-term savings and retirement accounts usually use?

Answer

They almost always use compound interest. Over decades, interest earning its own interest makes a massive difference in the final amount. That is exactly why saving for the long term is so powerful.

#468 · Financial Literacy

What is a budget?

Answer

A budget is a plan for your money over a set period, usually a month. It compares income (money coming in) to expenses (money going out). A good budget makes sure expenses do not add up to more than income.

#469 · Financial Literacy

What is income?

Answer

Income is money coming in, such as pay from a job, an allowance, or money from a small business. In a budget, income is the total you have available before any spending. Everything you spend or save has to fit inside it.

#470 · Financial Literacy

What is a fixed expense? Give examples.

Answer

A fixed expense stays the same amount every month. Examples are rent, a phone plan, or a monthly transit pass. Because they do not change, fixed expenses are the easiest part of a budget to predict.

#471 · Financial Literacy

What is a variable expense? Give examples.

Answer

A variable expense changes from month to month because it depends on your choices. Examples are groceries, entertainment, and eating out. Variable expenses are usually the easiest place to cut back when a budget is tight.

#472 · Financial Literacy

What is the difference between a need and a want?

Answer

A need is something you must have to live and function, like housing, food, and basic transportation. A want is nice to have but not required, like streaming subscriptions or eating out. A strong budget covers all needs first, then splits what is left between wants and savings.

#473 · Financial Literacy

Income \$2400. Fixed expenses \$1050, variable expenses \$550. How much is left over?

Answer

Add all expenses: $1050 + 550 = \$1600$. Subtract from income: $2400 - 1600 = \$800$. That \$800 is available for saving or extra spending, and the budget shows exactly how much there is.

#474 · Financial Literacy

Rent \$900, phone \$60, transit pass \$90. What are total fixed expenses?

Answer

Add them: $900 + 60 + 90 = \$1050$. All three count as fixed expenses because they cost the same amount every month. Grouping them makes a monthly budget quick to build.

#475 · Financial Literacy

Is a streaming subscription a need or a want?

Answer

It is a want. You can live and function without it, unlike housing, food, or basic transportation. Wants are the first place to cut when expenses are bigger than income.

#476 · Financial Literacy

What does it mean if your expenses are bigger than your income?

Answer

You are spending more than you earn, so you are either draining savings or going into debt. A budget exposes this problem clearly by putting income and expenses side by side. The fix is to cut expenses, starting with wants, or increase income.

#477 · Financial Literacy

Why does saving money early in life matter so much?

Answer

Compound interest needs time to work, and money saved earlier has more years to grow. Interest earns interest, so each extra year multiplies the effect. Starting even a few years earlier can lead to a much larger final amount, even with the same deposit and rate.

#478 · Financial Literacy

\$1000 invested at 6% compounded annually for 20 years. Final amount?

Answer

Use $A = 1000(1.06)^{20}$. Since 1.06^{20} is about 3.207135, the final amount is about \$3207.14. The money more than triples without adding a single extra dollar.

#479 · Financial Literacy

Amara invests \$1000 at 6% at age 15, Ben at age 25. Compare them at 35.

Answer

Amara gets 20 years of growth: $1000(1.06)^{20} = \$3207.14$. Ben gets only 10 years: $1000(1.06)^{10} = \$1790.85$. Amara ends up with \$1416.29 more, even though they invested the same amount at the same rate. The only difference was starting earlier.

#480 · Financial Literacy

Same deposit, same rate. What single factor made one investor finish with far more?

Answer

Time. Starting earlier gives compound interest more years to build on itself. This is why financial advisors say the best time to start saving is now, not later.

#481 · Financial Literacy

How is a credit card different from a debit card?

Answer

A credit card borrows money from a bank up to a limit, and you pay it back later. A debit card takes money directly out of an account you already own. Every credit card purchase is really a small loan.

#482 · Financial Literacy

When does a credit card charge you no interest?

Answer

Most credit cards charge no interest if you pay off your full balance every month. Interest only starts when you carry a balance, meaning you do not pay the full amount owed. Then the leftover amount gets charged a very high rate, often near 20% per year.

#483 · Financial Literacy

A card charges 20% per year, applied monthly. What is the monthly rate?

Answer

Divide the annual rate by 12: $20\% / 12$ is about 1.667% per month, or 0.01667 as a decimal. That monthly rate is what gets applied to any balance you carry. It looks small but adds up fast on a big balance.

#484 · Financial Literacy

Sam owes \$500 at 20% annual interest, charged monthly. Interest for one month?

Answer

The monthly rate is $0.20 / 12$, which is about 0.01667. Multiply: $500 \times 0.01667 = \$8.34$, rounded to the nearest cent. That \$8.34 gets added to what Sam owes if he carries the balance.

#485 · Financial Literacy

Sam owes \$500, monthly interest is \$8.34, and he pays only \$25. New balance?

Answer

Of the \$25 payment, \$8.34 just covers the interest, so only $25 - 8.34 = \$16.66$ reduces the debt. The new balance is $500 - 16.66 = \$483.34$. Even after paying \$25, the debt barely moved.

#486 · Financial Literacy

Why are minimum credit card payments dangerous?

Answer

A large chunk of each minimum payment just covers the interest, so the balance shrinks very slowly. Paying only the minimum can take years to clear even a small debt. The total interest paid can end up costing more than the original purchase.

#487 · Financial Literacy

What is a credit score?

Answer

A credit score is a number that shows how reliably a person pays back borrowed money. It rises when bills are paid on time and balances stay low, and it falls when payments are missed or balances stay high. A strong score makes future loans easier to get, often at better rates.

#488 · Financial Literacy

What is HST and what is the rate in Ontario?

Answer

HST stands for Harmonized Sales Tax, the sales tax added to the price of most things you buy in Ontario. The rate is 13%. That means every \$100 of taxable spending costs an extra \$13 in tax.

#489 · Financial Literacy

An item costs \$45 before tax. Find the total with 13% HST.

Answer

Tax = $45 \times 0.13 = \$5.85$. Total = $45 + 5.85 = \$50.85$. You can also do it in one step: $45 \times 1.13 = \$50.85$.

#490 · Financial Literacy

What is the one-step shortcut for adding 13% HST to a price?

Answer

Multiply the price by 1.13. That works because you are paying 100% of the price plus another 13%, and $100\% + 13\% = 113\% = 1.13$. For example, $60 \times 1.13 = \$67.80$.

#491 · Financial Literacy

An item costs \$80 before tax. How much HST do you pay?

Answer

Multiply the price by the tax rate: $80 \times 0.13 = \$10.40$. The total price would be $80 + 10.40 = \$90.40$. Check whether the question wants the tax alone or the full total.

#492 · Financial Literacy

A \$50 item is 20% off, then 13% HST is added. Final price?

Answer

Apply the discount first: $50 \times 0.80 = \$40$. Then add the tax: $40 \times 1.13 = \$45.20$. Read carefully which applies first, because doing the steps in the wrong order can change the total on some problems.

#493 · Financial Literacy

How is income tax different from sales tax?

Answer

Sales tax is added to the price of things you buy, like the 13% HST in Ontario. Income tax is a percentage taken from money a person earns, usually from a job. They are calculated on completely different amounts: one on spending, the other on earnings.

#494 · Financial Literacy

What does progressive taxation mean?

Answer

It means the tax rate increases as income increases. People who earn more pay a higher percentage, but only on the portion of income above certain thresholds, not on all of it at one flat rate. Lower amounts of income are always taxed at the lower rates.

#495 · Financial Literacy

True or false: in Canada, a high earner pays the top rate on ALL their income.

Answer

False. Canada's progressive system applies higher rates only to the portion of income above each threshold. The lower slices of everyone's income are taxed at the lower rates, no matter how much they earn in total.

#496 · Financial Literacy

A \$1000 phone: pay \$1000 now, or \$90 a month for 12 months. Which costs more?

Answer

The payment plan costs $90 \times 12 = \$1080$ in total, which is \$80 more than paying in full. That \$80 is effectively the interest for spreading the cost over a year. Always compare total cost, not just the monthly payment.

#497 · Financial Literacy

Why can a lower monthly payment still mean a more expensive loan?

Answer

A smaller payment usually means the loan lasts longer or carries a higher interest rate, so interest builds up over more months. The true cost is the total of every payment plus all fees, not the size of one payment. Multiply the payment by the number of payments to see the real price.

#498 · Financial Literacy

How do you find the true cost of borrowing on a payment plan?

Answer

Multiply the payment amount by the number of payments, then add any fees. Compare that total to the cash price of the item. The difference is what the borrowing actually costs you.

#499 · Financial Literacy

What is a payday loan and why is it risky?

Answer

A payday loan is a short-term loan meant to be repaid by your next paycheck. Its fees are extremely high and can work out to an annual interest rate of several hundred percent. Borrowers who cannot repay on time often take another loan, getting trapped in a costly cycle.

#500 · Financial Literacy

What is the hidden danger of buy-now-pay-later plans?

Answer

They split a purchase into several small payments, which feels harmless. But missing a payment can trigger steep fees or interest, and it is easy to lose track of several plans running at once. Small payments can quietly add up to more than your budget can handle.